1.30 Hyperbolic Functions

Worked Solutions

1 a)

$$2\sinh x \equiv 2(\tfrac{e^x - e^{-x}}{2})$$
$$\equiv e^x - e^{-x}$$

1 b)

$$8\cosh x \equiv 8(\tfrac{e^x + e^{-x}}{2})$$
$$\equiv 4e^x + 4e^{-x}$$

1 c)

$$4\sinh x - 4\cosh x \equiv 4(\tfrac{e^x - e^{-x}}{2}) - 4(\tfrac{e^x + e^{-x}}{2})$$
$$\equiv 2(e^x - e^{-x}) - 2(e^x + e^{-x})$$
$$\equiv 2e^x - 2e^{-x} - 2e^x - 2e^{-x}$$
$$\equiv -4e^{-x}$$

1 d)

$$5\cosh x - 10\sinh x \equiv 5(\tfrac{e^x + e^{-x}}{2}) - 10(\tfrac{e^x - e^{-x}}{2})$$
$$\equiv \frac{5}{2}(e^x + e^{-x}) - 5(e^x - e^{-x})$$
$$\equiv \frac{5}{2}e^x + \frac{5}{2}e^{-x} - 5e^x + 5e^{-x}$$
$$\equiv -\frac{5}{2}e^x + \frac{15}{2}e^{-x}$$

2 a)

$$2\sinh x + 4\cosh x \equiv 2(\tfrac{e^x - e^{-x}}{2}) + 4(\tfrac{e^x + e^{-x}}{2})$$
$$\equiv (e^x - e^{-x}) + 2(e^x + e^{-x})$$
$$\equiv e^x - e^{-x} + 2e^x + 2e^{-x}$$
$$\equiv 3e^x + e^{-x}$$
$$\equiv e^{-x}(3e^{2x} + 1)$$

2 b)

$$3\sinh x - 9\cosh x \equiv 3(\tfrac{e^x - e^{-x}}{2}) - 9(\tfrac{e^x + e^{-x}}{2})$$
$$\equiv \frac{3}{2}(e^x - e^{-x}) - \frac{9}{2}(e^x + e^{-x})$$
$$\equiv \frac{3}{2}e^x - \frac{3}{2}e^{-x} - \frac{9}{2}e^x - \frac{9}{2}e^{-x}$$
$$\equiv -3e^x - 6e^{-x}$$
$$\equiv -3e^{-x}(e^{2x} + 2)$$

2 c)

$$10\cosh x - 8\sinh x \equiv 10(\tfrac{e^x+e^{-x}}{2}) - 8(\tfrac{e^x-e^{-x}}{2})$$

$$\equiv 5(e^x + e^{-x}) - 4(e^x - e^{-x})$$

$$\equiv 5e^x + 5e^{-x} - 4e^x + 4e^{-x}$$

$$\equiv e^x + 9e^{-x}$$

$$\equiv e^{-x}(e^{2x} + 9)$$

2 d)

$$26\cosh x + 42\sinh x \equiv 26(\tfrac{e^x+e^{-x}}{2}) + 42(\tfrac{e^x-e^{-x}}{2})$$

$$\equiv 13(e^x + e^{-x}) + 21(e^x - e^{-x})$$

$$\equiv 13e^x + 13e^{-x} + 21e^x - 21e^{-x}$$

$$\equiv 34e^x - 8e^{-x}$$

$$\equiv 2e^{-x}(17e^{2x} - 4)$$

3)

$$55\sinh x - 35\cosh x \equiv 55(\tfrac{e^x-e^{-x}}{2}) - 35(\tfrac{e^x+e^{-x}}{2})$$

$$\equiv \frac{55}{2}(e^x - e^{-x}) - \frac{35}{2}(e^x + e^{-x})$$

$$\equiv \frac{55}{2}e^x - \frac{55}{2}e^{-x} - \frac{35}{2}e^x - \frac{35}{2}e^{-x}$$

$$\equiv 10e^x - 45e^{-x}$$

$$\equiv 5e^{-x}(2e^{2x} - 9)$$

4 a)

$\sinh 0 \equiv \frac{e^0-e^{-0}}{2} \equiv \frac{1-1}{2} = 0$

4 b)

$\cosh 0 = \frac{e^0+e^{-0}}{2} = \frac{1+1}{2} = 1$

5 a)

$\sinh 1 = \frac{e^1-e^{-1}}{2} = \frac{e-e^{-1}}{2} = 1.18$ (3 SF)

5 b)

$\cosh 1 = \frac{e^1+e^{-1}}{2} = \frac{e+e^{-1}}{2} = 1.54$ (3 SF)

5 c)

$\sinh(-1) = \frac{e^{-1}-e^{-(-1)}}{2} = \frac{e^{-1}-e}{2} = \frac{1/e-e}{2} = -1.18$ (3 SF)

5 d)

$\cosh(-1) = \frac{e^{-1}+e^{-(-1)}}{2} = \frac{e^{-1}+e}{2} = \frac{1/e+e}{2} = 1.54$ (3 SF)

6)

LHS: $\sinh(-x) \equiv \frac{e^{-x}-e^{x}}{2} \equiv -\frac{e^{x}-e^{-x}}{2} \equiv -\sinh x$: RHS

7)

LHS: $\cosh(-x) \equiv \frac{e^{-x}+e^{x}}{2} \equiv \frac{e^{x}+e^{-x}}{2} \equiv \cosh x$: RHS

8 a)

$\sinh(\ln 2) = \frac{e^{\ln 2}-e^{-\ln 2}}{2} = \frac{e^{\ln 2}-e^{\ln(2^{-1})}}{2} = \frac{2-2^{-1}}{2} = \frac{3}{4}$

8 b)

$\cosh(\ln 3) = \frac{e^{\ln 3}+e^{-\ln 3}}{2} = \frac{e^{\ln 3}+e^{\ln(3^{-1})}}{2} = \frac{3+3^{-1}}{2} = \frac{5}{3}$

8 c)

$\sinh(\ln \frac{1}{3}) = \frac{e^{\ln \frac{1}{3}}-e^{-\ln \frac{1}{3}}}{2} = \frac{e^{\ln \frac{1}{3}}-e^{\ln(\frac{1}{3})^{-1}}}{2} = \frac{\frac{1}{3}-3}{2} = -\frac{4}{3}$

8 d)

$\cosh(\ln \frac{1}{4}) = \frac{e^{\ln \frac{1}{4}}+e^{-\ln \frac{1}{4}}}{2} = \frac{e^{\ln \frac{1}{4}}+e^{\ln(\frac{1}{4})^{-1}}}{2} = \frac{\frac{1}{4}+4}{2} = \frac{17}{8}$

9)

$\sinh(\ln k) = \frac{e^{\ln k}-e^{-\ln k}}{2} = \frac{e^{\ln k}-e^{\ln k^{-1}}}{2} = \frac{k-k^{-1}}{2} = \frac{k^2-1}{2k}$

10)

$\cosh(2\ln k) = \frac{e^{2\ln k}+e^{-2\ln k}}{2} = \frac{e^{\ln k^2}+e^{\ln k^{-2}}}{2} = \frac{k^2+\frac{1}{k^2}}{2} = \frac{k^4+1}{2k^2}$

11)

$\tanh x \equiv \frac{\sinh x}{\cosh x} \equiv \frac{\frac{e^{x}-e^{-x}}{2}}{\frac{e^{x}+e^{-x}}{2}} \equiv \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}} \equiv \frac{e^{2x}-1}{e^{2x}+1} \equiv \frac{(e^{x}+1)(e^{x}-1)}{e^{2x}+1}$

12)

LHS: $\frac{1}{2}\ln(\frac{1+\tanh x}{1-\tanh x}) = \frac{1}{2}\ln(\frac{1+\frac{\sinh x}{\cosh x}}{1-\frac{\sinh x}{\cosh x}}) = \frac{1}{2}\ln(\frac{\cosh x+\sinh x}{\cosh x-\sinh x})$

$$= \frac{1}{2}\ln(\frac{\frac{e^{x}+e^{-x}}{2}+\frac{e^{x}-e^{-x}}{2}}{\frac{e^{x}+e^{-x}}{2}-\frac{e^{x}-e^{-x}}{2}}) = \frac{1}{2}\ln(\frac{e^{x}+e^{-x}+e^{x}-e^{-x}}{e^{x}+e^{-x}-(e^{x}-e^{-x})})$$

$$= \frac{1}{2}\ln(\frac{2e^{x}}{2e^{-x}}) = \frac{1}{2}\ln(\frac{e^{x}}{e^{-x}}) = \frac{1}{2}\ln(e^{x-(-x)}) = \frac{1}{2}\ln(e^{2x}) = \frac{1}{2}\times 2x = x\text{: RHS}$$

13 a)

$\tanh 0 = \frac{e^{2\times 0}-1}{e^{2\times 0}+1} = \frac{1-1}{1+1} = 0$

13 b)

$\tanh 1 = \frac{e^{2\times 1}-1}{e^{2\times 1}+1} = \frac{e^2-1}{e^2+1}$

13 c)

$\tanh(-1) = \frac{e^{2\times(-1)}-1}{e^{2\times(-1)}+1} = \frac{e^{-2}-1}{e^{-2}+1} = \frac{1-e^2}{1+e^2}$

14)

$\tanh(-x) \equiv \frac{e^{-2x}-1}{e^{-2x}+1} = \frac{1-e^{2x}}{1+e^{2x}} = -\frac{e^{2x}-1}{e^{2x}+1} = -\tanh x$

15 a)

$\tanh(\ln 3) = \frac{3-3^{-1}}{3+3^{-1}} = \frac{9-1}{9+1} = \frac{4}{5}$

15 b)

$\tanh(-\ln 4) = \frac{4^{-1}-4}{4^{-1}+4} = \frac{1-16}{1+16} = -\frac{15}{17}$

15 c)

$\tanh(\ln \frac{1}{5}) = \frac{\frac{1}{5}-5}{\frac{1}{5}+5} = \frac{1-25}{1+25} = -\frac{12}{13}$

16)

$\sinh x + \cosh x = \frac{e^x-e^{-x}}{2} + \frac{e^x+e^{-x}}{2} = e^x$

Since $e^x > 0$ for all real x, $\sinh x + \cosh x > 0$.

17)

$\sinh x - \cosh x = \frac{e^x-e^{-x}}{2} - \frac{e^x+e^{-x}}{2} = -e^{-x}$

Since $e^{-x} > 0$ for all real x, $\sinh x - \cosh x < 0$.

18 a)

$$\sinh x = 0 \Rightarrow \frac{e^x-e^{-x}}{2} = 0 \Rightarrow e^x - e^{-x} = 0$$
$$\Rightarrow e^{2x} - 1 = 0 \Rightarrow e^{2x} = 1 \Rightarrow 2x = 0 \Rightarrow x = 0$$

18 b)

$$\sinh x = 1 \Rightarrow \frac{e^x-e^{-x}}{2} = 1$$
$$\Rightarrow e^x - e^{-x} = 2$$
$$\Rightarrow e^{2x} - 1 = 2e^x$$
$$\Rightarrow e^{2x} - 2e^x - 1 = 0$$
$$\Rightarrow (e^x - 1)^2 - 1 - 1 = 0$$
$$\Rightarrow (e^x - 1)^2 = 2$$
$$\Rightarrow e^x = 1 \pm \sqrt{2}$$
$$\Rightarrow x = \ln(1+\sqrt{2}) \text{ or } x = \ln(1-\sqrt{2})$$

However, $1 - \sqrt{2} < 0$, so the only real solution is $x = \ln(1 + \sqrt{2})$.

18 c)

$\sinh x = -10 \Rightarrow \frac{e^x - e^{-x}}{2} = -10$

$$\Rightarrow e^x - e^{-x} = -20$$
$$\Rightarrow e^{2x} - 1 = -20e^x$$
$$\Rightarrow e^{2x} + 20e^x - 1 = 0$$
$$\Rightarrow (e^x + 10)^2 - 10^2 - 1 = 0$$
$$\Rightarrow (e^x + 10)^2 = 101$$
$$\Rightarrow e^x = -10 \pm \sqrt{101}$$
$$\Rightarrow x = \ln(-10 + \sqrt{101}) \text{ or } x = \ln(-10 - \sqrt{101})$$

However, $-10 - \sqrt{101} < 0$, so the only real solution is $x = \ln(-10 + \sqrt{101})$.

19 a)

$\cosh x = 1 \Rightarrow \frac{e^x + e^{-x}}{2} = 1$

$$\Rightarrow e^x + e^{-x} = 2$$
$$\Rightarrow e^{2x} + 1 = 2e^x$$
$$\Rightarrow e^{2x} - 2e^x + 1 = 0$$
$$\Rightarrow (e^x - 1)^2 - 1^2 + 1 = 0$$
$$\Rightarrow (e^x - 1)^2 = 0$$
$$\Rightarrow e^x = 1$$
$$\Rightarrow x = 0$$

19 b)

$\cosh x = 5 \Rightarrow \frac{e^x + e^{-x}}{2} = 5$

$$\Rightarrow e^x + e^{-x} = 10$$
$$\Rightarrow e^{2x} + 1 = 10e^x$$
$$\Rightarrow e^{2x} - 10e^x + 1 = 0$$
$$\Rightarrow (e^x - 5)^2 - 5^2 + 1 = 0$$
$$\Rightarrow (e^x - 5)^2 = 24$$
$$\Rightarrow e^x = 5 \pm 2\sqrt{6}$$
$$\Rightarrow x = \ln(5 + 2\sqrt{6}) \text{ or } x = \ln(5 - 2\sqrt{6})$$

19 c)

$\cosh x = 23 \Rightarrow \frac{e^x + e^{-x}}{2} = 23$

$$\Rightarrow e^x + e^{-x} = 46$$
$$\Rightarrow e^{2x} + 1 = 46e^x$$
$$\Rightarrow e^{2x} - 46e^x + 1 = 0$$
$$\Rightarrow (e^x - 23)^2 - 23^2 + 1 = 0$$
$$\Rightarrow (e^x - 23)^2 = 528$$
$$\Rightarrow e^x = 23 \pm 4\sqrt{33}$$
$$\Rightarrow x = \ln(23 + 4\sqrt{33}) \text{ or } x = \ln(23 - 4\sqrt{33})$$

20 a)

$\tanh x = 0 \Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = 0$

$$\Rightarrow e^{2x} - 1 = 0$$
$$\Rightarrow e^{2x} = 1$$
$$\Rightarrow 2x = 0$$
$$\Rightarrow x = 0$$

20 b)

$\tanh x = \frac{1}{3} \Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{1}{3}$

$$\Rightarrow 3e^{2x} - 3 = e^{2x} + 1$$
$$\Rightarrow 2e^{2x} = 4$$
$$\Rightarrow e^{2x} = 2$$
$$\Rightarrow 2x = \ln 2$$
$$\Rightarrow x = \frac{1}{2}\ln 2$$

20 c)

$\tanh x = -\frac{4}{5} \Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = -\frac{4}{5}$

$$\Rightarrow 5e^{2x} - 5 = -4e^{2x} - 4$$
$$\Rightarrow 9e^{2x} = 1$$
$$\Rightarrow e^{2x} = \frac{1}{9}$$
$$\Rightarrow 2x = \ln\frac{1}{9}$$
$$\Rightarrow x = \frac{1}{2}\ln\frac{1}{9}$$
$$\Rightarrow x = \ln(\frac{1}{9})^{1/2} = \ln\frac{1}{3} = -\ln 3$$

21 a)

$\sinh x + \cosh x = 2 \Rightarrow \frac{e^{x}-e^{-x}}{2} + \frac{e^{x}+e^{-x}}{2} = 2$

$$\Rightarrow e^{x} - e^{-x} + e^{x} + e^{-x} = 4$$
$$\Rightarrow 2e^{x} = 4$$
$$\Rightarrow e^{x} = 2$$
$$\Rightarrow x = \ln 2$$

21 b)

$2\sinh x - \cosh x = 1 \Rightarrow 2 \times \frac{e^x - e^{-x}}{2} - \frac{e^x + e^{-x}}{2} = 1$

$$\Rightarrow e^x - e^{-x} - \frac{e^x + e^{-x}}{2} = 1$$
$$\Rightarrow 2e^x - 2e^{-x} - e^x - e^{-x} = 2$$
$$\Rightarrow e^x - 3e^{-x} = 2$$
$$\Rightarrow e^{2x} - 3 = 2e^x$$
$$\Rightarrow e^{2x} - 2e^x - 3 = 0$$
$$\Rightarrow (e^x + 1)(e^x - 3) = 0$$
$$\Rightarrow e^x = -1 \text{ or } e^x = 3$$
$$\Rightarrow x = \ln 3 \text{ only as } e^x \neq -1$$

21 c)

$\sinh x - 2\cosh x = -2 \Rightarrow \frac{e^x - e^{-x}}{2} - 2 \times \frac{e^x + e^{-x}}{2} = -2$

$$\Rightarrow \frac{e^x - e^{-x}}{2} - (e^x + e^{-x}) = -2$$
$$\Rightarrow e^x - e^{-x} - 2e^x - 2e^{-x} = -4$$
$$\Rightarrow -e^x - 3e^{-x} = -4$$
$$\Rightarrow -e^{2x} - 3 = -4e^x$$
$$\Rightarrow e^{2x} - 4e^x + 3 = 0$$
$$\Rightarrow (e^x - 3)(e^x - 1) = 0$$
$$\Rightarrow e^x = 3 \text{ or } e^x = 1$$
$$\Rightarrow x = \ln 3 \text{ or } x = 0$$

22 a)

$2\cosh x - 3\sinh x = \frac{1}{2} \Rightarrow 2 \times \frac{e^x + e^{-x}}{2} - 3 \times \frac{e^x - e^{-x}}{2} = \frac{1}{2}$

$$\Rightarrow e^x + e^{-x} - \frac{3}{2}(e^x - e^{-x}) = \frac{1}{2}$$
$$\Rightarrow 2e^x + 2e^{-x} - 3e^x + 3e^{-x} = 1$$
$$\Rightarrow -e^x + 5e^{-x} = 1$$
$$\Rightarrow -e^{2x} + 5 = e^x$$
$$\Rightarrow e^{2x} + e^x - 5 = 0$$
$$\Rightarrow (e^x + \frac{1}{2})^2 - (\frac{1}{2})^2 - 5 = 0$$
$$\Rightarrow (e^x + \frac{1}{2})^2 = \frac{21}{4}$$
$$\Rightarrow e^x = -\frac{1}{2} \pm \frac{\sqrt{21}}{2}$$
$$\Rightarrow x = \ln(-\frac{1}{2} + \frac{\sqrt{21}}{2}) \text{ or } x = \ln(-\frac{1}{2} - \frac{\sqrt{21}}{2})$$

However, $-\frac{1}{2} - \frac{\sqrt{21}}{2} < 0$, so the only real solution is $x = \ln(-\frac{1}{2} + \frac{\sqrt{21}}{2})$.

22 b)

$5\sinh x - 6\cosh x = -7 \Rightarrow 5 \times \frac{e^x - e^{-x}}{2} - 6 \times \frac{e^x + e^{-x}}{2} = -7$

$$\Rightarrow 5(e^x - e^{-x}) - 6(e^x + e^{-x}) = -14$$

$$\Rightarrow 5e^x - 5e^{-x} - 6e^x - 6e^{-x} = -14$$

$$\Rightarrow -e^x - 11e^{-x} = -14$$

$$\Rightarrow -e^{2x} - 11 = -14e^x$$

$$\Rightarrow e^{2x} - 14e^x + 11 = 0$$

$$\Rightarrow (e^x - 7)^2 - 7^2 + 11 = 0$$

$$\Rightarrow (e^x - 7)^2 = 38$$

$$\Rightarrow e^x = 7 \pm \sqrt{38}$$

$$\Rightarrow x = \ln(7 + \sqrt{38}) \text{ or } x = \ln(7 - \sqrt{38})$$

22 c)

$3\cosh x - 5\sinh x = 2 \Rightarrow 3 \times \frac{e^x + e^{-x}}{2} - 5 \times \frac{e^x - e^{-x}}{2} = 2$

$$\Rightarrow 3(e^x + e^{-x}) - 5(e^x - e^{-x}) = 4$$

$$\Rightarrow 3e^x + 3e^{-x} - 5e^x + 5e^{-x} = 4$$

$$\Rightarrow -2e^x + 8e^{-x} = 4$$

$$\Rightarrow -e^x + 4e^{-x} = 2$$

$$\Rightarrow -e^{2x} + 4 = 2e^x$$

$$\Rightarrow e^{2x} + 2e^x - 4 = 0$$

$$\Rightarrow (e^x + 1)^2 - 1^2 - 4 = 0$$

$$\Rightarrow (e^x + 1)^2 = 5$$

$$\Rightarrow e^x = -1 \pm \sqrt{5}$$

$$\Rightarrow x = \ln(-1 + \sqrt{5}) \text{ or } x = \ln(-1 - \sqrt{5})$$

However, $-1 - \sqrt{5} < 0$, so the only real solution is $x = \ln(-1 + \sqrt{5})$.

22 d)

$4\sinh x + 3\cosh x = 4 \Rightarrow 4 \times \frac{e^x - e^{-x}}{2} + 3 \times \frac{e^x + e^{-x}}{2} = 4$

$$\Rightarrow 2e^x - 2e^{-x} + \frac{3}{2}(e^x + e^{-x}) = 4$$

$$\Rightarrow 4e^x - 4e^{-x} + 3e^x + 3e^{-x} = 8$$

$$\Rightarrow 7e^x - e^{-x} = 8$$

$$\Rightarrow 7e^{2x} - 1 = 8e^x$$

$$\Rightarrow 7e^{2x} - 8e^x - 1 = 0$$

$$\Rightarrow e^{2x} - \frac{8}{7}e^x - \frac{1}{7} = 0$$

$$\Rightarrow (e^x - \frac{4}{7})^2 - (\frac{4}{7})^2 - \frac{1}{7} = 0$$

$$\Rightarrow (e^x - \frac{4}{7})^2 = \frac{23}{49}$$

$$\Rightarrow e^x = \frac{4}{7} \pm \frac{\sqrt{23}}{7}$$

$$\Rightarrow x = \ln(\frac{4}{7} + \frac{\sqrt{23}}{7}) \text{ or } x = \ln(\frac{4}{7} - \frac{\sqrt{23}}{7})$$

However, $\frac{4}{7} - \frac{\sqrt{23}}{7} < 0$, so the only real solution is $x = \ln(\frac{4}{7} + \frac{\sqrt{23}}{7})$.

23 a)

$2\sinh 2x = 2 - 7e^{2x}$

$$\Rightarrow 2 \times \frac{e^{2x} - e^{-2x}}{2} = 2 - 7e^{2x}$$

$$\Rightarrow e^{2x} - e^{-2x} = 2 - 7e^{2x}$$

$$\Rightarrow e^{4x} - 1 = 2e^{2x} - 7e^{4x}$$

$$\Rightarrow 8e^{4x} - 2e^{2x} - 1 = 0$$

$$\Rightarrow (4e^{2x} + 1)(2e^{2x} - 1) = 0$$

$$\Rightarrow 4e^{2x} + 1 = 0 \text{ or } 2e^{2x} - 1 = 0$$

First consider $4e^{2x} + 1 = 0$:

As $e^{2x} > 0$, this equation has no real solutions.

Now consider $2e^{2x} - 1 = 0$:

$$\Rightarrow e^{2x} = \frac{1}{2} \Rightarrow 2x = \ln\frac{1}{2} \Rightarrow x = \frac{1}{2}\ln\frac{1}{2}$$

23 b)

$3\sinh 2x = 2e^{2x} - 2$

$$\Rightarrow 3 \times \frac{e^{2x} - e^{-2x}}{2} = 2e^{2x} - 2$$

$$\Rightarrow 3(e^{2x} - e^{-2x}) = 4e^{2x} - 4$$

$$\Rightarrow 3e^{2x} - 3e^{-2x} = 4e^{2x} - 4$$

$$\Rightarrow 3e^{4x} - 3 = 4e^{4x} - 4e^{2x}$$

$$\Rightarrow e^{4x} - 4e^{2x} + 3 = 0$$

$$\Rightarrow (e^{2x} - 3)(e^{2x} - 1) = 0$$

$$\Rightarrow e^{2x} - 3 = 0 \text{ or } e^{2x} - 1 = 0$$

First consider $e^{2x} - 3 = 0$:

$$\Rightarrow e^{2x} = 3 \Rightarrow 2x = \ln 3 \Rightarrow x = \frac{1}{2}\ln 3$$

Now consider $e^{2x} - 1 = 0$:

$$\Rightarrow e^{2x} = 1 \Rightarrow x = 0$$

$\therefore x = \frac{1}{2}\ln 3$ or $x = 0$.

24 a)

$(y^2 - 6y + 1)(y^2 + y + 1) \equiv y^4 + y^3 + y^2 - 6y^3 - 6y^2 - 6y + y^2 + y + 1$

$$\equiv y^4 - 5y^3 - 4y^2 - 5y + 1$$

24 b)

$\cosh 2x = 2 + 5\cosh x \Rightarrow \frac{e^{2x}+e^{-2x}}{2} = 2 + 5 \times \frac{e^x+e^{-x}}{2}$

$$\Rightarrow e^{2x} + e^{-2x} = 4 + 5(e^x + e^{-x})$$

$$\Rightarrow e^{2x} + e^{-2x} = 4 + 5e^x + 5e^{-x}$$

$$\Rightarrow e^{4x} + 1 = 4e^{2x} + 5e^{3x} + 5e^x$$

$$\Rightarrow e^{4x} - 5e^{3x} - 4e^{2x} - 5e^x + 1 = 0$$

$$\Rightarrow (e^{2x} - 6e^x + 1)(e^{2x} + e^x + 1) = 0$$

$$\Rightarrow e^{2x} - 6e^x + 1 = 0 \text{ or } e^{2x} + e^x + 1 = 0$$

First consider $e^{2x} + e^x + 1 = 0$:

As $e^{2x} > 0$ and $e^x > 0$, this equation has no real solutions.

Now consider $e^{2x} - 6e^x + 1 = 0$:

$$\Rightarrow (e^x - 3)^2 - 3^2 + 1 = 0$$

$$\Rightarrow (e^x - 3)^2 = 8$$

$$\Rightarrow e^x = 3 \pm 2\sqrt{2}$$

$$\Rightarrow x = \ln(3 + 2\sqrt{2}) \text{ or } x = \ln(3 - 2\sqrt{2})$$

25 a)

i) $f(1) = 1^4 - 4 \times 1^3 + 4 \times 1 - 1 = 1 - 4 + 4 - 1 = 0$

$$f(-1) = (-1)^4 - 4 \times (-1)^3 + 4 \times (-1) - 1 = 1 + 4 - 4 - 1 = 0$$

$\therefore$ both $(y - 1)$ and $(y + 1)$ are factors of $f(y)$ by the Factor Theorem.

ii) $f(y) = y^4 - 4y^3 + 4y - 1$

$$\equiv (y - 1)(y^3 - 3y^2 - 3y + 1)$$

$$\equiv (y - 1)(y + 1)(y^2 - 4y + 1)$$

25 b)

$\sinh 2x = 4\sinh x \Rightarrow \frac{e^{2x}-e^{-2x}}{2} = 4 \times \frac{e^x-e^{-x}}{2}$

$$\Rightarrow e^{2x} - e^{-2x} = 4(e^x - e^{-x})$$

$$\Rightarrow e^{2x} - e^{-2x} = 4e^x - 4e^{-x}$$

$$\Rightarrow e^{4x} - 1 = 4e^{3x} - 4e^x$$

$$\Rightarrow e^{4x} - 4e^{3x} + 4e^x - 1 = 0$$

$$\Rightarrow (e^x - 1)(e^x + 1)(e^{2x} - 4e^x + 1) = 0$$

$$\Rightarrow e^x - 1 = 0 \text{ or } e^x + 1 = 0 \text{ or } e^{2x} - 4e^x + 1 = 0$$

First consider $e^x + 1 = 0$:

As $e^x > 0$, this equation has no real solutions.

Now consider $e^x - 1 = 0$:

$$\Rightarrow e^x = 1 \Rightarrow x = 0$$

Now consider $e^{2x} - 4e^x + 1 = 0$:

$$\Rightarrow (e^x - 2)^2 - 2^2 + 1 = 0$$
$$\Rightarrow (e^x - 2)^2 = 3$$
$$\Rightarrow e^x = 2 \pm \sqrt{3}$$
$$\Rightarrow x = \ln(2 + \sqrt{3}) \text{ or } x = \ln(2 - \sqrt{3})$$

So the solutions to $\sinh 2x = 4 \sinh x$ are $x = 0$, or $x = \ln(2 + \sqrt{3})$ or $x = \ln(2 - \sqrt{3})$.

26 a)

$\sinh^2 x + \sinh x - 6 = 0$

$$\Rightarrow (\sinh x + 3)(\sinh x - 2) = 0$$
$$\Rightarrow \sinh x = -3 \text{ or } \sinh x = 2$$

First consider $\sinh x = -3$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = -3$$
$$\Rightarrow e^x - e^{-x} = -6$$
$$\Rightarrow e^{2x} - 1 = -6e^x$$
$$\Rightarrow e^{2x} + 6e^x - 1 = 0$$
$$\Rightarrow (e^x + 3)^2 - 3^2 - 1 = 0$$
$$\Rightarrow (e^x + 3)^2 = 10$$
$$\Rightarrow e^x = -3 \pm \sqrt{10}$$
$$\Rightarrow x = \ln(-3 + \sqrt{10}) \text{ or } x = \ln(-3 - \sqrt{10})$$

However, $-3 - \sqrt{10} < 0$, so the only real solution is $x = \ln(-3 + \sqrt{10})$.

Now consider $\sinh x = 2$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = 2$$
$$\Rightarrow e^x - e^{-x} = 4$$
$$\Rightarrow e^{2x} - 1 = 4e^x$$
$$\Rightarrow e^{2x} - 4e^x - 1 = 0$$
$$\Rightarrow (e^x - 2)^2 - 2^2 - 1 = 0$$
$$\Rightarrow (e^x - 2)^2 = 5$$
$$\Rightarrow e^x = 2 \pm \sqrt{5}$$
$$\Rightarrow x = \ln(2 + \sqrt{5}) \text{ or } x = \ln(2 - \sqrt{5})$$

However, $2 - \sqrt{5} < 0$, so the only real solution is $x = \ln(2 + \sqrt{5})$.

So $x = \ln(-3 + \sqrt{10})$ and $x = \ln(2 + \sqrt{5})$ solve the original equation.

26 b)

$8\cosh^2 x - 26\cosh x + 15 = 0$

$$\Rightarrow (4\cosh x - 3)(2\cosh x - 5) = 0$$

$$\Rightarrow \cosh x = \frac{3}{4} \text{ or } \cosh x = \frac{5}{2}$$

First consider $\cosh x = \frac{3}{4}$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = \frac{3}{4}$ has no real solutions.

Now consider $\cosh x = \frac{5}{2}$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = \frac{5}{2}$$

$$\Rightarrow e^x + e^{-x} = 5$$

$$\Rightarrow e^{2x} + 1 = 5e^x$$

$$\Rightarrow e^{2x} - 5e^x + 1 = 0$$

$$\Rightarrow (e^x - \frac{5}{2})^2 - (\frac{5}{2})^2 + 1 = 0$$

$$\Rightarrow (e^x - \frac{5}{2})^2 = \frac{21}{4}$$

$$\Rightarrow e^x = \frac{5}{2} \pm \frac{\sqrt{21}}{2}$$

$$\Rightarrow x = \ln(\frac{5}{2} + \frac{\sqrt{21}}{2}) \text{ or } x = \ln(\frac{5}{2} - \frac{\sqrt{21}}{2})$$

27 a)

$\cosh^3 x - 5\cosh^2 x - 2\cosh x + 24 = 0$

$$\Rightarrow (\cosh x + 2)(\cosh x - 3)(\cosh x - 4) = 0$$

$$\Rightarrow \cosh x = -2 \text{ or } \cosh x = 3 \text{ or } \cosh x = 4$$

First consider $\cosh x = -2$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -2$ has no real solutions.

Now consider $\cosh x = 3$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 3 \Rightarrow e^x + e^{-x} = 6$$

$$\Rightarrow e^{2x} + 1 = 6e^x \Rightarrow e^{2x} - 6e^x + 1 = 0$$

$$\Rightarrow (e^x - 3)^2 - 3^2 + 1 = 0 \Rightarrow (e^x - 3)^2 = 8$$

$$\Rightarrow e^x = 3 \pm 2\sqrt{2} \Rightarrow x = \ln(3 + 2\sqrt{2}) \text{ or } x = \ln(3 - 2\sqrt{2})$$

Now consider $\cosh x = 4$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 4 \Rightarrow e^x + e^{-x} = 8$$

$$\Rightarrow e^{2x} + 1 = 8e^x \Rightarrow e^{2x} - 8e^x + 1 = 0$$

$$\Rightarrow (e^x - 4)^2 - 4^2 + 1 = 0 \Rightarrow (e^x - 4)^2 = 15$$

$$\Rightarrow e^x = 4 \pm \sqrt{15} \Rightarrow x = \ln(4 + \sqrt{15}) \text{ or } x = \ln(4 - \sqrt{15})$$

So $x = \ln(3 \pm 2\sqrt{2})$ and $x = \ln(4 \pm \sqrt{15})$ solve the original equation.

27 b)

$12\cosh^4 x - 44\cosh^2 x - 45 = 0$

$$\Rightarrow (6\cosh^2 x + 5)(2\cosh^2 x - 9) = 0$$

$$\Rightarrow \cosh^2 x = -\frac{5}{6} \text{ or } \cosh^2 x = \frac{9}{2}$$

Firstly, $\cosh^2 x = -\frac{5}{6}$ cannot have any real solutions.

Now consider $\cosh^2 x = \frac{9}{2} \Rightarrow \cosh x = \pm\frac{3\sqrt{2}}{2}$.

First consider $\cosh x = -\frac{3\sqrt{2}}{2}$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -\frac{3\sqrt{2}}{2}$ has no real solutions.

Now consider $\cosh x = \frac{3\sqrt{2}}{2}$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = \frac{3\sqrt{2}}{2} \Rightarrow e^x + e^{-x} = 3\sqrt{2}$$

$$\Rightarrow e^{2x} + 1 = 3\sqrt{2}e^x \Rightarrow e^{2x} - 3\sqrt{2}e^x + 1 = 0$$

$$\Rightarrow (e^x - \frac{3\sqrt{2}}{2})^2 - (\frac{3\sqrt{2}}{2})^2 + 1 = 0$$

$$\Rightarrow (e^x - \frac{3\sqrt{2}}{2})^2 = \frac{7}{2}$$

$$\Rightarrow e^x = \frac{3\sqrt{2}}{2} \pm \frac{\sqrt{14}}{2}$$

$$\Rightarrow x = \ln(\frac{3\sqrt{2}}{2} + \frac{\sqrt{14}}{2}) \text{ or } x = \ln(\frac{3\sqrt{2}}{2} - \frac{\sqrt{14}}{2})$$

28 a)

$15\tanh^2 x - 8\tanh x + 1 = 0$

$$\Rightarrow (5\tanh x - 1)(3\tanh x - 1) = 0$$

$$\Rightarrow \tanh x = \frac{1}{5} \text{ or } \tanh x = \frac{1}{3}$$

First consider $\tanh x = \frac{1}{5}$:

$$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{1}{5} \Rightarrow 5(e^{2x} - 1) = e^{2x} + 1$$

$$\Rightarrow 5e^{2x} - 5 = e^{2x} + 1 \Rightarrow 4e^{2x} = 6 \Rightarrow e^{2x} = \frac{3}{2}$$

$$\Rightarrow 2x = \ln\frac{3}{2} \Rightarrow x = \frac{1}{2}\ln\frac{3}{2}$$

Now consider $\tanh x = \frac{1}{3}$:

$$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{1}{3} \Rightarrow 3(e^{2x} - 1) = e^{2x} + 1$$

$$\Rightarrow 3e^{2x} - 3 = e^{2x} + 1 \Rightarrow 2e^{2x} = 4 \Rightarrow e^{2x} = 2$$

$$\Rightarrow 2x = \ln 2 \Rightarrow x = \frac{1}{2}\ln 2$$

So $x = \frac{1}{2}\ln\frac{3}{2}$ and $x = \frac{1}{2}\ln 2$ solve the original equation.

28 b)

$12\tanh^2 x - 17\tanh x + 6 = 0$

$$\Rightarrow (3\tanh x - 2)(4\tanh x - 3) = 0$$

$$\Rightarrow \tanh x = \frac{2}{3} \text{ or } \tanh x = \frac{3}{4}$$

First consider $\tanh x = \frac{2}{3}$:

$$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{2}{3} \Rightarrow 3(e^{2x}-1) = 2(e^{2x}+1)$$

$$\Rightarrow 3e^{2x} - 3 = 2e^{2x} + 2 \Rightarrow e^{2x} = 5 \Rightarrow 2x = \ln 5 \Rightarrow x = \frac{1}{2}\ln 5$$

Now consider $\tanh x = \frac{3}{4}$:

$$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{3}{4} \Rightarrow 4(e^{2x}-1) = 3(e^{2x}+1)$$

$$\Rightarrow 4e^{2x} - 4 = 3e^{2x} + 3 \Rightarrow e^{2x} = 7 \Rightarrow 2x = \ln 7 \Rightarrow x = \frac{1}{2}\ln 7$$

So $x = \frac{1}{2}\ln 5$ and $x = \frac{1}{2}\ln 7$ solve the original equation.

29 a)

i) $f(1) = 1^3 - 7 \times 1^2 + 7 \times 1 - 1 = 1 - 7 + +7 - 1 = 0 \therefore$ both $(y-1)$ is a factor of $f(y)$ by the Factor Theorem.

ii) $f(y) = y^3 - 7y^3 + 7y - 1$

$$\equiv (y-1)(y^2 - 6y + 1)$$

29 b)

$3\tanh 2x = 4\tanh x$

$$\Rightarrow 3 \times \frac{e^{2(2x)}-1}{e^{2(2x)}+1} = 4 \times \frac{e^{2x}-1}{e^{2x}+1}$$

$$\Rightarrow 3(e^{4x}-1)(e^{2x}+1) = 4(e^{4x}+1)(e^{2x}-1)$$

$$\Rightarrow 3(e^{6x} + e^{4x} - e^{2x} - 1) = 4(e^{6x} - e^{4x} + e^{2x} - 1)$$

$$\Rightarrow 3e^{6x} + 3e^{4x} - 3e^{2x} - 3 = 4e^{6x} - 4e^{4x} + 4e^{2x} - 4$$

$$\Rightarrow 0 = e^{6x} - 7e^{4x} + 7e^{2x} - 1$$

$$\Rightarrow (e^{2x}-1)(e^{4x} - 6e^{2x} + 1) = 0$$

$$\Rightarrow e^{2x} = 1 \text{ or } e^{4x} - 6e^{2x} + 1 = 0$$

First consider $e^{2x} = 1$:

$$\Rightarrow 2x = 0 \Rightarrow x = 0$$

Now consider $e^{4x} - 6e^{2x} + 1 = 0$:

$$\Rightarrow (e^{2x} - 3)^2 - 3^2 + 1 = 0 \Rightarrow (e^{2x} - 3)^2 = 8$$

$$\Rightarrow e^{2x} = 3 \pm 2\sqrt{2}$$

$$\Rightarrow x = \frac{1}{2}\ln(3 + 2\sqrt{2}) \text{ or } x = \frac{1}{2}\ln(3 - 2\sqrt{2})$$

So $x = 0$, $x = \frac{1}{2}\ln(3+2\sqrt{2})$ and $x = \frac{1}{2}\ln(3-2\sqrt{2})$ solve the original equation.

$\frac{1}{2}\ln(3+2\sqrt{2}) = \ln((3+2\sqrt{2})^{1/2})$

$(a+b\sqrt{2})^2 = 3+2\sqrt{2}$, where a and b are rational numbers.

$$\Rightarrow a^2 + 2b^2 + 2ab\sqrt{2} = 3 + 2\sqrt{2}$$

$\therefore a^2 + 2b^2 = 3$ (1)

$$ab = 1 \Rightarrow b = \frac{1}{a} \quad (2)$$

Substitute (2) into (1):

$$a^2 + 2(\frac{1}{a})^2 = 3 \Rightarrow a^4 + 2 = 3a^2 \Rightarrow a^4 - 3a^2 + 2 = 0 \Rightarrow (a^2-2)(a^2-1) = 0$$

$a^2 - 2 = 0 \Rightarrow a = \pm\sqrt{2}$ but a is a rational number, so this is not a solution.

$a^2 - 1 = 0 \Rightarrow a = \pm 1$

When $a = 1$, $b = \frac{1}{1} = 1$

When $a = -1$, $b = \frac{1}{-1} = -1$

$(3+2\sqrt{2})^{1/2} > 0$ so $(3+2\sqrt{2})^{1/2} = 1+\sqrt{2}$

$\frac{1}{2}\ln(3-2\sqrt{2}) = \ln((3-2\sqrt{2})^{1/2})$

$(a+b\sqrt{2})^2 = 3-2\sqrt{2}$, where a and b are rational numbers.

$$\Rightarrow a^2 + 2b^2 + 2ab\sqrt{2} = 3 - 2\sqrt{2}$$

$\therefore a^2 + 2b^2 = 3$ (1)

$$ab = -1 \Rightarrow b = -\frac{1}{a} \quad (2)$$

Substitute (2) into (1):

$$a^2 + 2(-\frac{1}{a})^2 = 3 \Rightarrow a^4 + 2 = 3a^2 \Rightarrow a^4 - 3a^2 + 2 = 0 \Rightarrow (a^2-2)(a^2-1) = 0$$

$a^2 - 2 = 0 \Rightarrow a = \pm\sqrt{2}$ but a is a rational number, so this is not a solution.

$a^2 - 1 = 0 \Rightarrow a = \pm 1$

When $a = 1$, $b = -\frac{1}{1} = -1$

When $a = -1$, $b = -\frac{1}{-1} = 1$

$\ln(1-\sqrt{2})$ is undefined so $(3-2\sqrt{2})^{1/2} = -1+\sqrt{2}$ is the only solution in this case.

So $x = 0$, $x = \frac{1}{2}\ln(3+2\sqrt{2}) = \ln(1+\sqrt{2})$ and $x = \frac{1}{2}\ln(3-2\sqrt{2}) = \ln(\sqrt{2}-1)$ are the three real solutions of $3\tanh 2x = 4\tanh x$.

30 a)

$\frac{1}{\cosh x} + \frac{1}{\sinh x} = e^{3x}$

$$\Rightarrow \frac{\sinh x + \cosh x}{\sinh x \cosh x} = e^{3x}$$

$$\Rightarrow \sinh x + \cosh x = e^{3x}\sinh x \cosh x$$

$$\Rightarrow \frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2} = e^{3x} \times \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2}$$

$$\Rightarrow 2(e^x - e^{-x}) + 2(e^x + e^{-x}) = e^{3x}(e^x - e^{-x})(e^x + e^{-x})$$

$$\Rightarrow 2e^x - 2e^{-x} + 2e^x + 2e^{-x} = e^{3x}(e^{2x} - e^{-2x})$$
$$\Rightarrow 4e^x = e^{5x} - e^x \Rightarrow e^{5x} = 5e^x \Rightarrow e^{4x} = 5$$
$$\Rightarrow 4x = \ln 5 \Rightarrow x = \frac{1}{4}\ln 5$$

30 b)

$\frac{1}{\cosh x} - \frac{1}{\sinh x} = \frac{4}{e^{3x}}$

$$\Rightarrow \frac{\sinh x - \cosh x}{\sinh x \cosh x} = \frac{4}{e^{3x}}$$
$$\Rightarrow e^{3x}(\sinh x - \cosh x) = 4\sinh x \cosh x$$
$$\Rightarrow e^{3x}(\frac{e^x - e^{-x}}{2} - \frac{e^x + e^{-x}}{2}) = 4 \times \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2}$$
$$\Rightarrow 2e^{3x}(e^x - e^{-x} - (e^x + e^{-x})) = 4(e^x - e^{-x})(e^x + e^{-x})$$
$$\Rightarrow e^{3x}(-2e^{-x}) = 2(e^{2x} - e^{-2x})$$
$$\Rightarrow -2e^{2x} = 2e^{2x} - 2e^{-2x} \Rightarrow -2e^{4x} = 2e^{4x} - 2$$
$$\Rightarrow 4e^{4x} = 2 \Rightarrow e^{4x} = \frac{1}{2} \Rightarrow 4x = \ln\frac{1}{2} \Rightarrow x = \frac{1}{4}\ln\frac{1}{2}$$

30 c)

$\frac{1}{\cosh x} + \frac{\sinh x}{1} = \frac{2}{e^x}$

$$\Rightarrow \frac{1 + \sinh x \cosh x}{\cosh x} = \frac{2}{e^x}$$
$$\Rightarrow e^x(1 + \sinh x \cosh x) = 2\cosh x$$
$$\Rightarrow e^x(1 + \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2}) = 2 \times \frac{e^x + e^{-x}}{2}$$
$$\Rightarrow e^x(4 + (e^x - e^{-x})(e^x + e^{-x})) = 4(e^x + e^{-x})$$
$$\Rightarrow e^x(4 + e^{2x} - e^{-2x}) = 4e^x + 4e^{-x}$$
$$\Rightarrow 4e^x + e^{3x} - e^{-x} = 4e^x + 4e^{-x} \Rightarrow e^{3x} = 5e^{-x}$$
$$\Rightarrow e^{4x} = 5 \Rightarrow 4x = \ln 5 \Rightarrow x = \frac{1}{4}\ln 5$$

30 d)

$4\frac{\cosh x}{1} - \frac{1}{\sinh x} = \frac{2}{e^x}$

$$\Rightarrow \frac{4\sinh x \cosh x - 1}{\sinh x} = \frac{2}{e^x}$$
$$\Rightarrow e^x(4\sinh x \cosh x - 1) = 2\sinh x$$
$$\Rightarrow e^x(4 \times \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2} - 1) = 2 \times \frac{e^x - e^{-x}}{2}$$
$$\Rightarrow e^x((e^x - e^{-x})(e^x + e^{-x}) - 1) = e^x - e^{-x}$$
$$\Rightarrow e^x(e^{2x} - e^{-2x} - 1) = e^x - e^{-x}$$
$$\Rightarrow e^{3x} - e^{-x} - e^x = e^x - e^{-x} \Rightarrow e^{3x} - 2e^x = 0$$
$$\Rightarrow e^x(e^{2x} - 2) = 0$$

$e^x \neq 0$ so $e^{2x} - 2 = 0$

$$\Rightarrow e^{2x} = 2 \Rightarrow 2x = \ln 2 \Rightarrow x = \frac{1}{2}\ln 2$$

31 a)

i) $f(-1) = (-1)^4 + 8 \times (-1)^3 + 2 \times (-1)^2 - 16 \times (-1) - 11$

$$= 1 - 8 + 2 + 16 - 11 = 0$$

$\therefore (y+1)$ is a factor of $f(y)$ by the Factor Theorem.

ii) $f(y) = y^4 + 8y^3 + 2y^2 - 16y - 11$

$$\equiv (y+1)(y^3 + 7y^2 - 5y - 11)$$

$$\equiv (y+1)(y+1)(y^2 + 6y - 11)$$

31 b)

$\frac{2\sinh x}{1+\cosh x} + \frac{1+\cosh x}{2\sinh x} = 3$

$$\Rightarrow \frac{4\sinh^2 x + (1+\cosh x)^2}{2\sinh x(1+\cosh x)} = 3$$

$$\Rightarrow 4\sinh^2 x + (1+\cosh x)^2 = 6\sinh x(1+\cosh x)$$

$$\Rightarrow 4\sinh^2 x + 1 + 2\cosh x + \cosh^2 x = 6\sinh x + 6\sinh x \cosh x$$

$$\Rightarrow 4(\frac{e^x - e^{-x}}{2})^2 + 1 + 2 \times \frac{e^x + e^{-x}}{2} + (\frac{e^x + e^{-x}}{2})^2 = 6 \times \frac{e^x - e^{-x}}{2} + 6 \times \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2}$$

$$\Rightarrow (e^x - e^{-x})^2 + 1 + e^x + e^{-x} + \frac{1}{4}(e^x + e^{-x})^2 = 3(e^x - e^{-x}) + \frac{3}{2}(e^x - e^{-x})(e^x + e^{-x})$$

$$\Rightarrow e^{2x} - 2 + e^{-2x} + 1 + e^x + e^{-x} + \frac{1}{4}(e^{2x} + 2 + e^{-2x}) = 3e^x - 3e^{-x} + \frac{3}{2}(e^{2x} - e^{-2x})$$

$$\Rightarrow 4e^{2x} - 8 + 4e^{-2x} + 4 + 4e^x + 4e^{-x} + e^{2x} + 2 + e^{-2x} = 12e^x - 12e^{-x} + 6e^{2x} - 6e^{-2x}$$

$$\Rightarrow -2 = 8e^x - 16e^{-x} + e^{2x} - 11e^{-2x}$$

$$\Rightarrow -2e^{2x} = 8e^{3x} - 16e^x + e^{4x} - 11$$

$$\Rightarrow e^{4x} + 8e^{3x} + 2e^{2x} - 16e^x - 11 = 0$$

$$\Rightarrow (e^x + 1)^2(e^{2x} + 6e^x - 11) = 0$$

$$\Rightarrow e^x + 1 = 0 \text{ or } e^{2x} + 6e^x - 11 = 0$$

First consider $e^x + 1 = 0$:

$e^x > 0$ so this has no real solutions.

Now consider $e^{2x} + 6e^x - 11 = 0$:

$$(e^x + 3)^2 - 3^2 - 11 = 0 \Rightarrow (e^x + 3)^2 = 20$$

$$\Rightarrow e^x = -3 \pm 2\sqrt{5}$$

$$\Rightarrow x = \ln(-3 + 2\sqrt{5}) \text{ or } x = \ln(-3 - 2\sqrt{5})$$

However, $-3 - 2\sqrt{5} < 0$, so the only real solution is $x = \ln(-3 + 2\sqrt{5})$.

32)

$\cosh^3 x + \sinh 3x$

$$\equiv (\frac{e^x + e^{-x}}{2})^3 + \frac{e^{3x} - e^{-3x}}{2}$$
$$\equiv \frac{1}{8}(e^x + e^{-x})^3 + \frac{1}{2}(e^{3x} - e^{-3x})$$
$$\equiv \frac{1}{8}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x}) + \frac{1}{8}(4e^{3x} - 4e^{-3x})$$
$$\equiv \frac{1}{8}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x} + 4e^{3x} - 4e^{-3x})$$
$$\equiv \frac{1}{8}(5e^{3x} + 3e^x + 3e^{-x} - 3e^{-3x})$$

33 a)

i) $4\sinh^3 x - 4\cosh^3 x$

$$\equiv 4(\frac{e^x - e^{-x}}{2})^3 - 4(\frac{e^x + e^{-x}}{2})^3$$
$$\equiv \frac{1}{2}(e^x - e^{-x})^3 - \frac{1}{2}(e^x + e^{-x})^3$$
$$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}) - \frac{1}{2}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$$
$$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x} - (e^{3x} + 3e^x + 3e^{-x} + e^{-3x}))$$
$$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x} - e^{3x} - 3e^x - 3e^{-x} - e^{-3x})$$
$$\equiv \frac{1}{2}(-6e^x - 2e^{-3x}) \equiv -(e^{-3x} + 3e^x)$$

ii) $4\sinh^3 x + 4\cosh^3 x$

$$\equiv 4(\frac{e^x - e^{-x}}{2})^3 + 4(\frac{e^x + e^{-x}}{2})^3$$
$$\equiv \frac{1}{2}(e^x - e^{-x})^3 + \frac{1}{2}(e^x + e^{-x})^3$$
$$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}) + \frac{1}{2}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$$
$$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x} + e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$$
$$\equiv \frac{1}{2}(2e^{3x} + 6e^{-x}) \equiv e^{3x} + 3e^{-x}$$

33 b)

$16\sinh^6 x - 16\cosh^6 x$

$$\equiv (4\sinh^3 x + 4\cosh^3 x)(4\sinh^3 x - 4\cosh^3 x)$$
$$\equiv (e^{3x} + 3e^{-x})(-(e^{-3x} + 3e^x))$$
$$\equiv -(e^{3x} + 3e^{-x})(e^{-3x} + 3e^x)$$
$$\equiv -(1 + 3e^{4x} + 3e^{-4x} + 3)$$
$$\equiv -(3(e^{4x} + e^{-4x}) + 4)$$
$$\equiv -(6 \times \frac{e^{4x} + e^{-4x}}{2} + 4)$$
$$\equiv -2(3\cosh 4x + 2)$$

34)

LHS: $(\sinh x + \cosh x)^3 \equiv (\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2})^3$

$$\equiv (\frac{e^x - e^{-x} + e^x + e^{-x}}{2})^3$$

$$\equiv (\frac{2e^x}{2})^3 \equiv (e^x)^3 = e^{3x}$$

35 a)

LHS: $4\cosh^3 x \equiv 4(\frac{e^x + e^{-x}}{2})^3$

$$\equiv 4 \times \frac{(e^x + e^{-x})^3}{8}$$

$$\equiv \frac{1}{2}(e^x + e^{-x})^3$$

$$\equiv \frac{1}{2}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$$

$$\equiv \frac{e^{3x} + e^{-3x}}{2} + 3 \times \frac{e^x + e^{-x}}{2}$$

$$\equiv \cosh 3x + 3\cosh x : \text{ RHS}$$

35 b)

Let $y = \cosh x$ $4\cosh^3 x = \cosh 3x + 3\cosh x \Rightarrow 4y^3 = \cosh 3x + 3y$

$$\Rightarrow 4y^3 - 3y = \cosh 3x$$

$\cosh 3x = \frac{5}{4}$

$$\Rightarrow \frac{e^{3x} + e^{-3x}}{2} = \frac{5}{4} \Rightarrow e^{3x} + e^{-3x} = \frac{5}{2}$$

$$\Rightarrow e^{6x} + 1 = \frac{5}{2}e^{3x} \Rightarrow e^{6x} - \frac{5}{2}e^{3x} + 1 = 0$$

$$\Rightarrow (e^{3x} - \frac{5}{4})^2 - (\frac{5}{4})^2 + 1 = 0 \Rightarrow (e^{3x} - \frac{5}{4})^2 = \frac{9}{16}$$

$$\Rightarrow e^{3x} = \frac{5}{4} \pm \frac{3}{4} \Rightarrow x = \frac{1}{3}\ln(\frac{5}{4} \pm \frac{3}{4})$$

$$\Rightarrow x = \frac{1}{3}\ln 2 \text{ or } x = \frac{1}{3}\ln\frac{1}{2}$$

So $y = \cosh(\frac{1}{3}\ln 2)$ or $y = \cosh(\frac{1}{3}\ln\frac{1}{2})$. Consider $y = \cosh(\frac{1}{3}\ln 2)$ first:

$$\Rightarrow y = \frac{e^{\frac{1}{3}\ln 2} + e^{-\frac{1}{3}\ln 2}}{2} \Rightarrow y = \frac{e^{\ln 2^{1/3}} + e^{\ln 2^{-1/3}}}{2}$$

$$\Rightarrow y = \frac{2^{1/3} + 2^{-1/3}}{2} = 2^{-2/3} + 2^{-4/3}$$

Now consider $y = \cosh(\frac{1}{3}\ln\frac{1}{2})$:

$$\Rightarrow y = \frac{e^{\frac{1}{3}\ln\frac{1}{2}} + e^{-\frac{1}{3}\ln\frac{1}{2}}}{2} \Rightarrow y = \frac{e^{\frac{1}{3}\ln 2^{-1}} + e^{-\frac{1}{3}\ln 2^{-1}}}{2}$$

$$\Rightarrow y = \frac{e^{\ln 2^{-1/3}} + e^{\ln 2^{1/3}}}{2} \Rightarrow y = 2^{-4/3} + 2^{-2/3} \text{ which is the same result as before.}$$

36 a)

$\cosh x \tanh x \equiv \cosh x \times \frac{\sinh x}{\cosh x} \equiv \sinh x$

36 b)

$\frac{\tanh x}{\sinh x} \equiv \frac{\frac{\sinh x}{\cosh x}}{\sinh x} \equiv \frac{1}{\cosh x}$

35 c)

$$\frac{\tanh x}{\cosh x} \equiv \frac{\sinh x}{\cosh x}\frac{1}{\cosh x} \equiv \frac{\sinh x}{\cosh^2 x}$$

37 a)

$\sinh x \cosh x \tanh x \equiv \sinh x \cosh x \times \frac{\sinh x}{\cosh x} \equiv \sinh^2 x$

37 b)

$\sinh^2 x \cosh x \tanh^2 x \equiv \sinh^2 x \cosh x \times \frac{\sinh^2 x}{\cosh^2 x} \equiv \frac{\sinh^4 x}{\cosh x}$

37 c)

$\frac{\sinh x}{\cosh^2 x} \tanh x \equiv \frac{\sinh x}{\cosh^2 x} \times \frac{\sinh x}{\cosh x} \equiv \frac{\sinh^2 x}{\cosh^3 x} \equiv \frac{1}{\cosh x}$

37 d)

$\frac{\tanh^2 x}{\sinh^2 x} \cosh x \equiv \frac{\frac{\sinh^2 x}{\cosh^2 x}}{\sinh^2 x} \cosh x \equiv \frac{1}{\cosh^2 x} \cosh x \equiv \frac{1}{\cosh^3 x}$

38)

LHS: $\cosh^2 x - \sinh^2 x \equiv (\frac{e^x+e^{-x}}{2})^2 - (\frac{e^x-e^{-x}}{2})^2$

$$\equiv \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4}$$

$$\equiv \frac{e^{2x} + 2 + e^{-2x} - (e^{2x} - 2 + e^{-2x})}{4} \equiv \frac{4}{4} \equiv 1 : \text{ RHS}$$

39)

RHS: $2\sinh x \cosh x \equiv 2 \times \frac{e^x-e^{-x}}{2} \times \frac{e^x+e^{-x}}{2}$

$$\equiv \frac{(e^x - e^{-x})(e^x + e^{-x})}{2} \equiv \frac{e^{2x} - e^{-2x}}{2} \equiv \sinh 2x : \text{ LHS}$$

40 a)

RHS: $\cosh^2 x + \sinh^2 x \equiv (\frac{e^x+e^{-x}}{2})^2 + (\frac{e^x-e^{-x}}{2})^2$

$$\equiv \frac{(e^x + e^{-x})^2}{4} + \frac{(e^x - e^{-x})^2}{4}$$

$$\equiv \frac{e^{2x} + 2 + e^{-2x} + e^{2x} - 2 + e^{-2x}}{4} \equiv \frac{2e^{2x} + 2e^{-2x}}{4} \equiv \frac{e^{2x} + e^{-2x}}{2} \equiv \cosh 2x : \text{ LHS}$$

40 b)

RHS: $2\cosh^2 x - 1 \equiv 2(\frac{e^x+e^{-x}}{2})^2 - 1$

$$\equiv \frac{e^{2x}+2+e^{-2x}}{2} - 1 \equiv \frac{e^{2x}+e^{-2x}}{2} \equiv \cosh 2x: \text{ LHS}$$

40 c)

RHS: $1 + 2\sinh^2 x \equiv 1 + 2(\frac{e^x-e^{-x}}{2})^2$

$$\equiv 1 + \frac{e^{2x}-2+e^{-2x}}{2} \equiv \frac{e^{2x}+e^{-2x}}{2} \equiv \cosh 2x: \text{ LHS}$$

41)

RHS: $\frac{2\tanh x}{1+\tanh^2 x} \equiv \frac{2\times\frac{e^{2x}-1}{e^{2x}+1}}{1+(\frac{e^{2x}-1}{e^{2x}+1})^2}$

$$\equiv \frac{2(e^{2x}-1)(e^{2x}+1)}{(e^{2x}+1)^2+(e^{2x}-1)^2} \equiv \frac{2(e^{4x}-1)}{2e^{4x}+2} \equiv \frac{e^{4x}-1}{e^{4x}+1}$$

$$\equiv \frac{e^{2(2x)}-1}{e^{2(2x)}+1} \equiv \tanh 2x: \text{ LHS}$$

42 a)

$5\cosh^2 x + 3\sinh^2 x \equiv 5(1+\sinh^2 x) + 3\sinh^2 x$

$$\equiv 5 + 5\sinh^2 x + 3\sinh^2 x \equiv 5 + 8\sinh^2 x$$

42 b)

$2\sinh^2 x - 6\cosh^2 x \equiv 2\sinh^2 x - 6(1+\sinh^2 x)$

$$\equiv 2\sinh^2 x - 6 - 6\sinh^2 x \equiv -4\sinh^2 x - 6$$

43 a)

$9\cosh^2 x + 2\sinh^2 x \equiv 9\cosh^2 x + 2(\cosh^2 x - 1)$

$$\equiv 9\cosh^2 x + 2\cosh^2 x - 2 \equiv 11\cosh^2 x - 2$$

43 b)

$5\sinh^2 x - 8\cosh^2 x \equiv 5(\cosh^2 x - 1) - 8\cosh^2 x$

$$\equiv 5\cosh^2 x - 5 - 8\cosh^2 x \equiv -3\cosh^2 x - 5$$

44 a)

$\cosh 2x \equiv 2\cosh^2 x - 1 \Rightarrow \cosh^2 x \equiv \frac{\cosh 2x+1}{2}$ $\cosh 2x \equiv 1 + 2\sinh^2 x \Rightarrow \sinh^2 x \equiv \frac{\cosh 2x-1}{2}$ So:

$$2\cosh^2 x + 5\sinh^2 x \equiv 2 \times \frac{\cosh 2x+1}{2} + 5 \times \frac{\cosh 2x - 1}{2}$$

$$\equiv \cosh 2x + 1 + \frac{5}{2}\cosh 2x - \frac{5}{2} \equiv \frac{7}{2}\cosh 2x - \frac{3}{2}$$

44 b)

$8\sinh^2 x - 3\cosh^2 x \equiv 8 \times \frac{\cosh 2x - 1}{2} - 3 \times \frac{\cosh 2x + 1}{2}$

$$\equiv 4(\cosh 2x - 1) - \frac{3}{2}(\cosh 2x + 1) \equiv 4\cosh 2x - 4 - \frac{3}{2}\cosh 2x - \frac{3}{2} \equiv \frac{5}{2}\cosh 2x - \frac{11}{2}$$

45 a)

$\cosh^2 x + \sinh^2 x = 3$

$$\Rightarrow \cosh^2 x + \cosh^2 x - 1 = 3 \Rightarrow 2\cosh^2 x = 4 \Rightarrow \cosh^2 x = 2 \Rightarrow \cosh x = \pm\sqrt{2}$$

Consider $\cosh x = -\sqrt{2}$ first: $\cosh x \geq 1$ so the equation $\cosh x = -\sqrt{2}$ has no real solutions. Now consider $\cosh x = \sqrt{2}$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = \sqrt{2} \Rightarrow e^x + e^{-x} = 2\sqrt{2} \Rightarrow e^{2x} + 1 = 2\sqrt{2}e^x$$

$$\Rightarrow e^{2x} - 2\sqrt{2}e^x + 1 = 0 \Rightarrow (e^x - \sqrt{2})^2 - (\sqrt{2})^2 + 1 = 0$$

$$\Rightarrow (e^x - \sqrt{2})^2 = 1 \Rightarrow e^x = \sqrt{2} \pm 1 \Rightarrow x = \ln(\sqrt{2} \pm 1)$$

$$\Rightarrow x = \ln(\sqrt{2} + 1) \text{ or } x = \ln(\sqrt{2} - 1)$$

45 b)

$2\cosh^2 x + 3\sinh x - 4 = 0$

$$\Rightarrow 2(1 + \sinh^2 x) + 3\sinh x - 4 = 0 \Rightarrow 2 + 2\sinh^2 x + 3\sinh x - 4 = 0$$

$$\Rightarrow 2\sinh^2 x + 3\sinh x - 2 = 0 \Rightarrow (2\sinh x - 1)(\sinh x + 2) = 0$$

$$\Rightarrow \sinh x = \frac{1}{2} \text{ or } \sinh x = -2$$

Consider $\sinh x = \frac{1}{2}$ first:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{1}{2} \Rightarrow e^x - e^{-x} = 1 \Rightarrow e^{2x} - 1 = e^x \Rightarrow e^{2x} - e^x - 1 = 0$$

$$\Rightarrow (e^x - \frac{1}{2})^2 - (\frac{1}{2})^2 - 1 = 0 \Rightarrow (e^x - \frac{1}{2})^2 = \frac{5}{4}$$

$$\Rightarrow e^x = \frac{1}{2} \pm \frac{\sqrt{5}}{2} \Rightarrow x = \ln(\frac{1}{2} \pm \frac{\sqrt{5}}{2})$$

$$\Rightarrow x = \ln(\frac{1}{2} + \frac{\sqrt{5}}{2}) \text{ or } x = \ln(\frac{1}{2} - \frac{\sqrt{5}}{2})$$

However, $\frac{1}{2} - \frac{\sqrt{5}}{2} < 0$, so the only real solution is $x = \ln(\frac{1}{2} + \frac{\sqrt{5}}{2})$. Now consider $\sinh x = -2$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = -2 \Rightarrow e^x - e^{-x} = -4 \Rightarrow e^{2x} - 1 = -4e^x \Rightarrow e^{2x} + 4e^x - 1 = 0$$

$$\Rightarrow (e^x + 2)^2 - 2^2 - 1 = 0 \Rightarrow (e^x + 2)^2 = 5 \Rightarrow e^x = -2 \pm \sqrt{5}$$

$$\Rightarrow x = \ln(-2 + \sqrt{5}) \text{ or } x = \ln(-2 - \sqrt{5})$$

However, $-2 - \sqrt{5} < 0$, so the only real solution is $x = \ln(-2 + \sqrt{5})$. So $x = \ln(\frac{1}{2} + \frac{\sqrt{5}}{2})$ and $x = \ln(-2 + \sqrt{5})$ solve the original equation.

45 c)

$5\sinh x = 2\cosh^2 x$

$$\Rightarrow 5\sinh x = 2 \times (1 + \sinh^2 x) \Rightarrow 5\sinh x = 2 + 2\sinh^2 x$$

$$\Rightarrow 2\sinh^2 x - 5\sinh x + 2 = 0 \Rightarrow (2\sinh x - 1)(\sinh x - 2) = 0$$

$$\Rightarrow \sinh x = \frac{1}{2} \text{ or } \sinh x = 2$$

Consider $\sinh x = \frac{1}{2}$ first:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{1}{2} \Rightarrow e^x - e^{-x} = 1 \Rightarrow e^{2x} - 1 = e^x \Rightarrow e^{2x} - e^x - 1 = 0$$

$$\Rightarrow (e^x - \frac{1}{2})^2 - (\frac{1}{2})^2 - 1 = 0 \Rightarrow (e^x - \frac{1}{2})^2 = \frac{5}{4}$$

$$\Rightarrow e^x = \frac{1}{2} \pm \frac{\sqrt{5}}{2} \Rightarrow x = \ln(\frac{1}{2} \pm \frac{\sqrt{5}}{2})$$

$$\Rightarrow x = \ln(\frac{1}{2} + \frac{\sqrt{5}}{2}) \text{ or } x = \ln(\frac{1}{2} - \frac{\sqrt{5}}{2})$$

However, $\frac{1}{2} - \frac{\sqrt{5}}{2} < 0$, so the only real solution is $x = \ln(\frac{1}{2} + \frac{\sqrt{5}}{2})$. Now consider $\sinh x = 2$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = 2 \Rightarrow e^x - e^{-x} = 4 \Rightarrow e^{2x} - 1 = 4e^x \Rightarrow e^{2x} - 4e^x - 1 = 0$$

$$\Rightarrow (e^x - 2)^2 - 2^2 - 1 = 0 \Rightarrow (e^x - 2)^2 = 5 \Rightarrow e^x = 2 \pm \sqrt{5}$$

$$\Rightarrow x = \ln(2 + \sqrt{5}) \text{ or } x = \ln(2 - \sqrt{5})$$

However, $2 - \sqrt{5} < 0$, so the only real solution is $x = \ln(2 + \sqrt{5})$. So $x = \ln(\frac{1}{2} + \frac{\sqrt{5}}{2})$ and $x = \ln(2 + \sqrt{5})$ solve the original equation.

45 d)

$3\sinh^2 x = 2\cosh x + 5$

$$\Rightarrow 3(\cosh^2 x - 1) = 2\cosh x + 5 \Rightarrow 3\cosh^2 x - 3 = 2\cosh x + 5$$

$$\Rightarrow 3\cosh^2 x - 2\cosh x - 8 = 0 \Rightarrow (3\cosh x + 4)(\cosh x - 2) = 0$$

$$\Rightarrow \cosh x = -\frac{4}{3} \text{ or } \cosh x = 2$$

Consider $\cosh x = -\frac{4}{3}$ first: $\cosh x \geq 1$ for all real values of x, so this has no real solutions. Now consider $\cosh x = 2$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = 2 \Rightarrow e^x - e^{-x} = 4 \Rightarrow e^{2x} - 1 = 4e^x \Rightarrow e^{2x} - 4e^x - 1 = 0$$

$$\Rightarrow (e^x - 2)^2 - 2^2 - 1 = 0 \Rightarrow (e^x - 2)^2 = 5 \Rightarrow e^x = 2 \pm \sqrt{5}$$

$$\Rightarrow x = \ln(2 \pm \sqrt{5}) \Rightarrow x = \ln(2 + \sqrt{5}) \text{ or } x = \ln(2 - \sqrt{5})$$

However, $20 - \sqrt{5} < 0$, so the only real solution is $x = \ln(2 + \sqrt{5})$.

46 a)

$\cosh 2x = 3 + 7\cosh x$

$$\Rightarrow 2\cosh^2 x - 1 = 3 + 7\cosh x \Rightarrow 2\cosh^2 x - 7\cosh x - 4 = 0$$

$$\Rightarrow (2\cosh x + 1)(\cosh x - 4) = 0 \Rightarrow \cosh x = -\frac{1}{2} \text{ or } \cosh x = 4$$

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -\frac{1}{2}$ has no real solutions. Now consider $\cosh x = 4$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 4 \Rightarrow e^x + e^{-x} = 8 \Rightarrow e^{2x} + 1 = 8e^x \Rightarrow e^{2x} - 8e^x + 1 = 0$$

$$\Rightarrow (e^x - 4)^2 - 4^2 + 1 = 0 \Rightarrow (e^x - 4)^2 = 15 \Rightarrow e^x = 4 \pm \sqrt{15}$$

$$\Rightarrow x = \ln(4 \pm \sqrt{15}) \Rightarrow x = \ln(4 + \sqrt{15}) \text{ or } x = \ln(4 - \sqrt{15})$$

46 b)

$5\cosh 2x + 8\sinh x = 7$

$$\Rightarrow 5(1 + 2\sinh^2 x) + 8\sinh x = 7 \Rightarrow 5 + 10\sinh^2 x + 8\sinh x = 7$$

$$\Rightarrow 10\sinh^2 x + 8\sinh x - 2 = 0 \Rightarrow 5\sinh^2 x + 4\sinh x - 1 = 0$$

$$\Rightarrow (5\sinh x - 1)(\sinh x + 1) = 0 \Rightarrow \sinh x = \frac{1}{5} \text{ or } \sinh x = -1$$

First consider $\sinh x = \frac{1}{5}$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{1}{5} \Rightarrow e^x - e^{-x} = \frac{2}{5} \Rightarrow e^{2x} - 1 = \frac{2}{5}e^x$$

$$\Rightarrow e^{2x} - \frac{2}{5}e^x - 1 = 0 \Rightarrow (e^x - \frac{1}{5})^2 - (\frac{1}{5})^2 - 1 = 0$$

$$\Rightarrow (e^x - \frac{1}{5})^2 = \frac{26}{25} \Rightarrow e^x = \frac{1}{5} \pm \frac{\sqrt{26}}{5}$$

$$\Rightarrow x = \ln(\frac{1}{5} \pm \frac{\sqrt{26}}{5}) \Rightarrow x = \ln(\frac{1}{5} + \frac{\sqrt{26}}{5}) \text{ or } x = \ln(\frac{1}{5} - \frac{\sqrt{26}}{5})$$

However, $\frac{1}{5} - \frac{\sqrt{26}}{5} < 0$, so the only real solution is $x = \ln(\frac{1}{5} + \frac{\sqrt{26}}{5})$. Now consider $\sinh x = -1$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = -1 \Rightarrow e^x - e^{-x} = -2 \Rightarrow e^{2x} - 1 = -2e^x \Rightarrow e^{2x} + 2e^x - 1 = 0$$

$$\Rightarrow (e^x + 1)^2 - 1^2 - 1 = 0 \Rightarrow (e^x + 1)^2 = 2 \Rightarrow e^x = -1 \pm \sqrt{2}$$

$$\Rightarrow x = \ln(-1 \pm \sqrt{2}) \Rightarrow x = \ln(-1 + \sqrt{2}) \text{ or } x = \ln(-1 - \sqrt{2})$$

However, $-1 - \sqrt{2} < 0$, so the only real solution is $x = \ln(-1 + \sqrt{2})$. So $x = \ln(\frac{1}{5} + \frac{\sqrt{26}}{5})$ and $x = \ln(-1 + \sqrt{2})$ solve the original equation.

46 c)

$8\cosh 2x = \cosh x + 9$

$$\Rightarrow 8(2\cosh^2 x - 1) = \cosh x + 9 \Rightarrow 16\cosh^2 x - 8 = \cosh x + 9$$

$$\Rightarrow 16\cosh^2 x - \cosh x - 17 = 0 \Rightarrow (16\cosh x - 17)(\cosh x + 1) = 0$$

$$\Rightarrow \cosh x = \frac{17}{16} \text{ or } \cosh x = -1$$

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -1$ has no real solutions. Now consider $\cosh x = \frac{17}{16}$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = \frac{17}{16} \Rightarrow e^x + e^{-x} = \frac{17}{8} \Rightarrow e^{2x} + 1 = \frac{17}{8}e^x$$

$$\Rightarrow e^{2x} - \frac{17}{8}e^x + 1 = 0 \Rightarrow (e^x - \frac{17}{16})^2 - (\frac{17}{16})^2 + 1 = 0$$

$$\Rightarrow (e^x - \frac{17}{16})^2 = \frac{33}{256} \Rightarrow e^x = \frac{17}{16} \pm \frac{\sqrt{33}}{16}$$

$$\Rightarrow x = \ln(\frac{17}{16} \pm \frac{\sqrt{33}}{16}) \Rightarrow x = \ln(\frac{17}{16} + \frac{\sqrt{33}}{16}) \text{ or } x = \ln(\frac{17}{16} - \frac{\sqrt{33}}{16})$$

46 d)

$\sinh x = 5 - 2\cosh 2x$

$$\Rightarrow \sinh x = 5 - 2(1 + 2\sinh^2 x) \Rightarrow \sinh x = 5 - 2 - 4\sinh^2 x$$

$$\Rightarrow 4\sinh^2 x + \sinh x - 3 = 0 \Rightarrow (4\sinh x - 3)(\sinh x + 1) = 0 \Rightarrow \sinh x = \frac{3}{4} \text{ or } \sinh x = -1$$

First consider $\sinh x = \frac{3}{4}$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{3}{4} \Rightarrow e^x - e^{-x} = \frac{3}{2} \Rightarrow e^{2x} - 1 = \frac{3}{2}e^x$$

$$\Rightarrow e^{2x} - \frac{3}{2}e^x - 1 = 0 \Rightarrow (e^x - \frac{3}{4})^2 - (\frac{3}{4})^2 - 1 = 0 \Rightarrow (e^x - \frac{3}{4})^2 = \frac{25}{16}$$

$$\Rightarrow e^x = \frac{3}{4} \pm \frac{5}{4} \Rightarrow x = \ln(\frac{3}{4} \pm \frac{5}{4})$$

However, $\frac{3}{4} - \frac{5}{4} = -\frac{1}{2} < 0$, so the only real solution is $x = \ln(\frac{3}{4} + \frac{5}{4}) = \ln 2$. Now consider $\sinh x = -1$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = -1 \Rightarrow e^x - e^{-x} = -2 \Rightarrow e^{2x} - 1 = -2e^x \Rightarrow e^{2x} + 2e^x - 1 = 0$$

$$\Rightarrow (e^x + 1)^2 - 1^2 - 1 = 0 \Rightarrow (e^x + 1)^2 = 2 \Rightarrow e^x = -1 \pm \sqrt{2}$$

$$\Rightarrow x = \ln(-1 \pm \sqrt{2}) \Rightarrow x = \ln(-1 + \sqrt{2}) \text{ or } x = \ln(-1 - \sqrt{2})$$

However, $-1-\sqrt{2} < 0$, so the only real solution is $x = \ln(-1+\sqrt{2})$. So $x = \ln 2$ and $x = \ln(-1+\sqrt{2})$ solve the original equation.

47 a)

$\sinh 2x = 20\sinh x$

$$\Rightarrow 2\sinh x\cosh x = 20\sinh x \Rightarrow \sinh x\cosh x = 10\sinh x$$

$$\Rightarrow \sinh x\cosh x - 10\sinh x = 0 \Rightarrow \sinh x(\cosh x - 10) = 0 \Rightarrow \sinh x = 0 \text{ or } \cosh x = 10$$

First consider $\sinh x = 0$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = 0 \Rightarrow e^x - e^{-x} = 0 \Rightarrow e^{2x} - 1 = 0 \Rightarrow e^{2x} = 1 \Rightarrow 2x = 0 \Rightarrow x = 0$$

Now consider $\cosh x = 10$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 10 \Rightarrow e^x + e^{-x} = 20 \Rightarrow e^{2x} + 1 = 20e^x \Rightarrow e^{2x} - 20e^x + 1 = 0$$

$$\Rightarrow (e^x - 10)^2 - 10^2 + 1 = 0 \Rightarrow (e^x - 10)^2 = 99 \Rightarrow e^x = 10 \pm 3\sqrt{11}$$

$$\Rightarrow x = \ln(10 \pm 3\sqrt{11}) \Rightarrow x = \ln(10 + 3\sqrt{11}) \text{ or } x = \ln(10 - 3\sqrt{11})$$

So $x = 0$, $x = \ln(10 + 3\sqrt{11})$ and $x = \ln(10 - 3\sqrt{11})$ solve the original equation.

47 b)

$\frac{2}{3}\sinh 2x = \cosh x$

$$\Rightarrow \frac{2}{3} \times 2\sinh x\cosh x = \cosh x \Rightarrow \frac{4}{3}\sinh x\cosh x = \cosh x$$

$$\Rightarrow \frac{4}{3}\sinh x\cosh x - \cosh x = 0 \Rightarrow \cosh x(\frac{4}{3}\sinh x - 1) = 0$$

$$\Rightarrow \cosh x = 0 \text{ or } \sinh x = \frac{3}{4}$$

$\cosh x \geq 1$ for all real values of x, so $\cosh x = 0$ has no real solutions. Now consider $\sinh x = \frac{3}{4}$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{3}{4} \Rightarrow e^x - e^{-x} = \frac{3}{2} \Rightarrow e^{2x} - 1 = \frac{3}{2}e^x$$

$$\Rightarrow e^{2x} - \frac{3}{2}e^x - 1 = 0 \Rightarrow (e^x - \frac{3}{4})^2 - (\frac{3}{4})^2 - 1 = 0$$

$$\Rightarrow (e^x - \frac{3}{4})^2 = \frac{25}{16} \Rightarrow e^x = \frac{3}{4} \pm \frac{5}{4} \Rightarrow x = \ln(\frac{3}{4} \pm \frac{5}{4})$$

However, $\frac{3}{4} - \frac{5}{4} = -\frac{1}{2} < 0$, so the only real solution is $x = \ln(\frac{3}{4} + \frac{5}{4}) = \ln 2$.

48 a)

The first mistake is that Dylan substitutes $\cosh^2 x$ for $1 - \sinh^2 x$, when it should be substituted for $1 + \sinh^2 x$. The second mistake is on the 3rd line, where $+4$ has become -4 when attempting to complete the square.

48 b)

$3\sinh x + \cosh^2 x = 5$

$$\Rightarrow 3\sinh x + 1 + \sinh^2 x = 5 \Rightarrow \sinh^2 x + 3\sinh x - 4 = 0 \Rightarrow (\sinh x + 4)(\sinh x - 1) = 0$$

First consider $\sinh x = -4$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = -4 \Rightarrow e^x - e^{-x} = -8 \Rightarrow e^{2x} - 1 = -8e^x \Rightarrow e^{2x} + 8e^x - 1 = 0$$

$$\Rightarrow e^x = -4 \pm \sqrt{17} \Rightarrow x = \ln(-4 \pm \sqrt{17})$$

However, $-4 - \sqrt{17} < 0$, so the only real solution is $x = \ln(-4 + \sqrt{17})$. Now consider $\sinh x = 1$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = 1 \Rightarrow e^x - e^{-x} = 2 \Rightarrow e^{2x} - 1 = 2e^x \Rightarrow e^{2x} - 2e^x - 1 = 0$$

$$\Rightarrow e^x = 1 \pm \sqrt{2} \Rightarrow x = \ln(1 \pm \sqrt{2})$$

However, $1 - \sqrt{2} < 0$, so the only real solution is $x = \ln(1 + \sqrt{2})$. So $x = \ln(-4 + \sqrt{17})$ and $x = \ln(1 + \sqrt{2})$ solve the original equation.

49 a)

$\cosh 2x \equiv 2\cosh^2 x - 1 \Rightarrow \cosh^2 x \equiv \frac{\cosh 2x+1}{2} \therefore \cosh^2 2x \equiv \frac{\cosh 4x+1}{2}$ LHS: $8\cosh^4 x \equiv 8(\cosh^2 x)^2 \equiv 8(\frac{\cosh 2x+1}{2})^2$

$$\equiv 2(\cosh 2x + 1)^2 \equiv 2(\cosh^2 2x + 2\cosh 2x + 1)$$

$$\equiv 2(\frac{\cosh 4x + 1}{2} + 2\cosh 2x + 1) \equiv \cosh 4x + 1 + 4\cosh 2x + 2$$

$$\equiv \cosh 4x + 4\cosh 2x + 3 : \text{ RHS}$$

49 b)

$8\cosh^4 x \equiv \cosh 4x + 4\cosh 2x + 3$

$$\Rightarrow \cosh 4x \equiv 8\cosh^4 x - 4\cosh 2x - 3$$

$$\Rightarrow \cosh 4x \equiv 8\cosh^4 x - 4(2\cosh^2 x - 1) - 3 \equiv 8\cosh^4 x - 8\cosh^2 x + 1$$

Then: $2\cosh 4x - 23\cosh 2x - 33 = 0$

$$2(8\cosh^4 x - 8\cosh^2 x + 1) - 23(2\cosh^2 x - 1) - 33 = 0$$

$$\Rightarrow 16\cosh^4 x - 16\cosh^2 x + 2 - 46\cosh^2 x + 23 - 33 = 0$$

$$\Rightarrow 16\cosh^4 x - 62\cosh^2 x - 8 = 0 \Rightarrow 8\cosh^4 x - 31\cosh^2 x - 4 = 0$$

$$\Rightarrow (8\cosh^2 x + 1)(\cosh^2 x - 4) = 0 \Rightarrow \cosh^2 x = -\frac{1}{8} \text{ or } \cosh^2 x = 4$$

First consider $\cosh^2 x = -\frac{1}{8}$: $\cosh^2 x \geq 1$ for all real values of x, so this has no real solutions. Now consider $\cosh^2 x = 4 \Rightarrow \cosh x = \pm 2$. Firstly, $\cosh x = -2$: $\cosh x \geq 1$ for all real values of x, so this has no real solutions. Secondly, $\cosh x = 2$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 2 \Rightarrow e^x + e^{-x} = 4 \Rightarrow e^{2x} + 1 = 4e^x \Rightarrow e^{2x} - 4e^x + 1 = 0$$

$$\Rightarrow (e^x - 2)^2 - 2^2 + 1 = 0 \Rightarrow (e^x - 2)^2 = 3 \Rightarrow e^x = 2 \pm \sqrt{3}$$

$$\Rightarrow x = \ln(2 + \sqrt{3}) \text{ or } x = \ln(2 - \sqrt{3})$$

50 a)

$\sinh x(\sinh x + \cosh x) = 4$

$$\Rightarrow \frac{e^x - e^{-x}}{2} \times (\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2}) = 4$$

$$\Rightarrow \frac{e^x - e^{-x}}{2} \times \frac{2e^x}{2} = 4 \Rightarrow \frac{e^x - e^{-x}}{2} \times e^x = 4$$

$$\Rightarrow (e^x - e^{-x})e^x = 8 \Rightarrow e^{2x} - 1 = 8 \Rightarrow e^{2x} = 9 \Rightarrow 2x = \ln 9 \Rightarrow x = \frac{1}{2}\ln 9 = \ln 3$$

50 b)

$\cosh x(\sinh x + \cosh x) = 13$

$$\Rightarrow \frac{e^x + e^{-x}}{2} \times (\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2}) = 13$$

$$\Rightarrow \frac{e^x + e^{-x}}{2} \times \frac{2e^x}{2} = 13 \Rightarrow \frac{e^x + e^{-x}}{2} \times e^x = 13$$

$$\Rightarrow (e^x + e^{-x})e^x = 26 \Rightarrow e^{2x} + 1 = 26 \Rightarrow e^{2x} = 25 \Rightarrow 2x = \ln 25 \Rightarrow x = \frac{1}{2}\ln 25 = \ln 5$$

50 c)

$\sinh x(5\sinh x - 4\cosh x) = 3$

$$\Rightarrow \frac{e^x - e^{-x}}{2} \times (5 \times \frac{e^x - e^{-x}}{2} - 4 \times \frac{e^x + e^{-x}}{2}) = 3$$

$$\Rightarrow \frac{e^x - e^{-x}}{2} \times \frac{5(e^x - e^{-x}) - 4(e^x + e^{-x})}{2} = 3$$

$$\Rightarrow \frac{e^x - e^{-x}}{2} \times \frac{e^x - 9e^{-x}}{2} = 3 \Rightarrow (e^x - e^{-x})(e^x - 9e^{-x}) = 12$$

$$\Rightarrow e^{2x} - 10 + 9e^{-2x} = 12 \Rightarrow e^{4x} - 10e^{2x} + 9 = 12e^{2x}$$

$$\Rightarrow e^{4x} - 22e^{2x} + 9 = 0 \Rightarrow (e^{2x} - 11)^2 - 11^2 + 9 = 0$$

$$\Rightarrow (e^{2x} - 11)^2 = 112 \Rightarrow e^{2x} = 11 \pm 4\sqrt{7} \Rightarrow 2x = \ln(11 \pm 4\sqrt{7})$$

$$\Rightarrow x = \frac{1}{2}\ln(11 + 4\sqrt{7}) \text{ or } x = \frac{1}{2}\ln(11 - 4\sqrt{7})$$

51 a)

$3\tanh 2x - 4\tanh x = 0$

$$\Rightarrow 3 \times \frac{2\tanh x}{1 + \tanh^2 x} - 4\tanh x = 0 \Rightarrow 6\tanh x - 4\tanh x(1 + \tanh^2 x) = 0$$

$$\Rightarrow 6\tanh x - 4\tanh x - 4\tanh^3 x = 0 \Rightarrow 2\tanh x - 4\tanh^3 x = 0$$

$$\Rightarrow \tanh x - 2\tanh^3 x = 0 \Rightarrow \tanh x(1 - 2\tanh^2 x) = 0 \Rightarrow \tanh x = 0 \text{ or } \tanh^2 x = \frac{1}{2}$$

First consider $\tanh x = 0$:

$$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = 0 \Rightarrow e^{2x} - 1 = 0 \Rightarrow e^{2x} = 1 \Rightarrow 2x = 0 \Rightarrow x = 0$$

Now consider $\tanh^2 x = \frac{1}{2} \Rightarrow \tanh x = \pm\frac{\sqrt{2}}{2}$. Firstly $\tanh x = \frac{\sqrt{2}}{2}$:

$$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{\sqrt{2}}{2} \Rightarrow 2(e^{2x} - 1) = \sqrt{2}(e^{2x} + 1)$$

$$\Rightarrow 2e^{2x} - 2 = \sqrt{2}e^{2x} + \sqrt{2} \Rightarrow (2 - \sqrt{2})e^{2x} = 2 + \sqrt{2}$$

$$\Rightarrow e^{2x} = \frac{2 + \sqrt{2}}{2 - \sqrt{2}} = 3 + 2\sqrt{2} \Rightarrow 2x = \ln(3 + 2\sqrt{2}) \Rightarrow x = \frac{1}{2}\ln(3 + 2\sqrt{2})$$

Secondly $\tanh x = -\frac{\sqrt{2}}{2}$:

$$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = -\frac{\sqrt{2}}{2} \Rightarrow 2(e^{2x} - 1) = -\sqrt{2}(e^{2x} + 1)$$

$$\Rightarrow 2e^{2x} - 2 = -\sqrt{2}e^{2x} - \sqrt{2} \Rightarrow (2+\sqrt{2})e^{2x} = 2 - \sqrt{2}$$

$$\Rightarrow e^{2x} = \frac{2-\sqrt{2}}{2+\sqrt{2}} = 3 - 2\sqrt{2} \Rightarrow 2x = \ln(3-2\sqrt{2}) \Rightarrow x = \frac{1}{2}\ln(3-2\sqrt{2})$$

So $x = \frac{1}{2}\ln(3+2\sqrt{2})$ and $x = \frac{1}{2}\ln(3-2\sqrt{2})$ solve the original equation.

51 b)

$10\tanh 2x = 9 + \tanh x$

$$\Rightarrow 10 \times \frac{2\tanh x}{1+\tanh^2 x} = 9 + \tanh x \Rightarrow 20\tanh x = (9+\tanh x)(1+\tanh^2 x)$$

$$\Rightarrow 20\tanh x = 9 + 9\tanh^2 x + \tanh x + \tanh^3 x$$

$$\Rightarrow \tanh^3 x + 9\tanh^2 x - 19\tanh x + 9 = 0 \Rightarrow (\tanh x - 1)(\tanh^2 x + 10\tanh x - 9) = 0$$

$$\Rightarrow \tanh x = 1 \text{ or } \tanh^2 x + 10\tanh x - 9 = 0$$

First consider $\tanh x = 1$: $-1 < \tanh x < 1$ for all real values of x, so this has no real solutions. Now consider $\tanh^2 x + 10\tanh x - 9 = 0$:

$$\Rightarrow (\tanh x + 5)^2 - 5^2 - 9 = 0 \Rightarrow (\tanh x + 5)^2 = 34 \Rightarrow \tanh x = -5 \pm \sqrt{34}$$

$$\Rightarrow \tanh x = -5 + \sqrt{34} \text{ or } \tanh x = -5 - \sqrt{34}$$

Firstly, $\tanh x = -5 - \sqrt{34}$: $-1 < \tanh x < 1$ for all real values of x, so this has no real solutions. Secondly, $\tanh x = -5 + \sqrt{34}$:

$$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = -5 + \sqrt{34} \Rightarrow e^{2x} - 1 = (-5+\sqrt{34})(e^{2x}+1)$$

$$\Rightarrow e^{2x} - 1 = (-5+\sqrt{34})e^{2x} + (-5+\sqrt{34})$$

$$\Rightarrow (1 - (-5+\sqrt{34}))e^{2x} = -4 + \sqrt{34} \Rightarrow (6-\sqrt{34})e^{2x} = -4 + \sqrt{34}$$

$$\Rightarrow e^{2x} = \frac{-4+\sqrt{34}}{6-\sqrt{34}} = 5 + \sqrt{34} \Rightarrow 2x = \ln(5+\sqrt{34}) \Rightarrow x = \frac{1}{2}\ln(5+\sqrt{34})$$

52)

LHS: $\sinh x \cosh y + \cosh x \sinh y$

$$\equiv \frac{e^x - e^{-x}}{2} \times \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \times \frac{e^y - e^{-y}}{2}$$

$$\equiv \frac{(e^x - e^{-x})(e^y + e^{-y}) + (e^x + e^{-x})(e^y - e^{-y})}{4}$$

$$\equiv \frac{e^x e^y + e^x e^{-y} - e^{-x}e^y - e^{-x}e^{-y} + e^x e^y - e^x e^{-y} + e^{-x}e^y - e^{-x}e^{-y}}{4}$$

$$\equiv \frac{2e^x e^y - 2e^{-x}e^{-y}}{4} \equiv \frac{e^x e^y - e^{-x}e^{-y}}{2}$$

$$\equiv \frac{e^{x+y} - e^{-x-y}}{2} \equiv \frac{e^{x+y} - e^{-(x+y)}}{2} \equiv \sinh(x+y) : \text{ RHS}$$

53)

LHS: $\cosh x \cosh y - \sinh x \sinh y$

$$\equiv \frac{e^x + e^{-x}}{2} \times \frac{e^y + e^{-y}}{2} - \frac{e^x - e^{-x}}{2} \times \frac{e^y - e^{-y}}{2}$$

$$\equiv \frac{(e^x + e^{-x})(e^y + e^{-y}) - (e^x - e^{-x})(e^y - e^{-y})}{4}$$

$$\equiv \frac{e^x e^y + e^x e^{-y} + e^{-x} e^y + e^{-x} e^{-y} - (e^x e^y - e^x e^{-y} - e^{-x} e^y + e^{-x} e^{-y})}{4}$$

$$\equiv \frac{e^x e^y + e^x e^{-y} + e^{-x} e^y + e^{-x} e^{-y} - e^x e^y + e^x e^{-y} + e^{-x} e^y - e^{-x} e^{-y}}{4}$$

$$\equiv \frac{2e^x e^{-y} + 2e^{-x} e^y}{4} \equiv \frac{e^{x-y} + e^{-x+y}}{2} \equiv \frac{e^{x-y} + e^{-(x-y)}}{2} \equiv \cosh(x-y) : \text{ RHS}$$

54 a)

Using the compound angle formulae proved in Q48: $\sinh(x + \ln 5) = \cosh x$

$$\Rightarrow \sinh x \cosh(\ln 5) + \cosh x \sinh(\ln 5) = \cosh x$$

$$\Rightarrow \sinh x \times \frac{e^{\ln 5} + e^{-\ln 5}}{2} + \cosh x \times \frac{e^{\ln 5} - e^{-\ln 5}}{2} = \cosh x$$

$$\Rightarrow \sinh x \times \frac{5 + \frac{1}{5}}{2} + \cosh x \times \frac{5 - \frac{1}{5}}{2} = \cosh x$$

$$\Rightarrow \frac{13}{5} \sinh x + \frac{12}{5} \cosh x = \cosh x \Rightarrow \frac{13}{5} \sinh x + \frac{7}{5} \cosh x = 0$$

$$\Rightarrow 13 \sinh x + 7 \cosh x = 0 \Rightarrow 13 \sinh x = -7 \cosh x \Rightarrow \frac{\sinh x}{\cosh x} = -\frac{7}{13} \Rightarrow \tanh x = -\frac{7}{13} \text{ as required}$$

54 b)

$\frac{e^{2x}-1}{e^{2x}+1} = -\frac{7}{13}$

$$\Rightarrow 13(e^{2x} - 1) = -7(e^{2x} + 1) \Rightarrow 13e^{2x} - 13 = -7e^{2x} - 7$$

$$\Rightarrow 20e^{2x} = 6 \Rightarrow e^{2x} = \frac{3}{10} \Rightarrow 2x = \ln \frac{3}{10}$$

$$\Rightarrow x = \frac{1}{2} \ln \frac{3}{10} = \frac{1}{2} \ln(\frac{3}{2 \times 5}) = \frac{1}{2}(\ln 3 - \ln(2 \times 5))$$

$$= \frac{1}{2}(\ln 3 - (\ln 2 + \ln 5)) = \frac{1}{3}(\ln 3 - \ln 2 - \ln 5)$$

55)

LHS: $\sinh 3x \equiv \sinh(2x + x) \equiv \sinh 2x \cosh x + \cosh 2x \sinh x$

$$\equiv (2 \sinh x \cosh x) \cosh x + (1 + 2 \sinh^2 x) \sinh x$$

$$\equiv 2 \sinh x \cosh^2 x + \sinh x + 2 \sinh^3 x$$

$$\equiv 2 \sinh x (1 + \sinh^2 x) + \sinh x + 2 \sinh^3 x$$

$$\equiv 2 \sinh x + 2 \sinh^3 x + \sinh x + 2 \sinh^3 x \equiv 4 \sinh^3 x + 3 \sinh x : \text{ RHS}$$

56)

LHS: $\cosh 3x \equiv \cosh(2x+x) \equiv \cosh 2x \cosh x + \sinh 2x \sinh x$

$$\equiv (2\cosh^2 x - 1)\cosh x + (2\sinh x \cosh x)\sinh x$$

$$\equiv 2\cosh^3 x - \cosh x + 2\sinh^2 x \cosh x$$

$$\equiv 2\cosh^3 x - \cosh x + 2(\cosh^2 x - 1)\cosh x$$

$$\equiv 2\cosh^3 x - \cosh x + 2\cosh^3 x - 2\cosh x \equiv 4\cosh^3 x - 3\cosh x : \text{ RHS}$$

57 a)

$\sinh 3x = 15\sinh x \Rightarrow 4\sinh^3 x + 3\sinh x = 15\sinh x$

$$\Rightarrow 4\sinh^3 x - 12\sinh x = 0 \Rightarrow \sinh^3 x - 3\sinh x = 0 \Rightarrow \sinh x(\sinh^2 x - 3) = 0$$

$$\Rightarrow \sinh x = 0 \text{ or } \sinh x = \pm\sqrt{3}$$

If $\sinh x = 0 \Rightarrow x = 0$. If $\sinh x = \sqrt{3}$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = \sqrt{3} \Rightarrow e^x - e^{-x} = 2\sqrt{3} \Rightarrow e^{2x} - 1 = 2\sqrt{3}e^x$$

$$\Rightarrow e^{2x} - 2\sqrt{3}e^x - 1 = 0 \Rightarrow (e^x - \sqrt{3})^2 - 3 - 1 = 0 \Rightarrow (e^x - \sqrt{3})^2 = 4$$

$$\Rightarrow e^x = \sqrt{3} \pm 2 \Rightarrow x = \ln(\sqrt{3}+2) \text{ only as } \sqrt{3} - 2 < 0$$

If $\sinh x = -\sqrt{3}$:

$$\Rightarrow \frac{e^x - e^{-x}}{2} = -\sqrt{3} \Rightarrow e^x - e^{-x} = -2\sqrt{3} \Rightarrow e^{2x} - 1 = -2\sqrt{3}e^x$$

$$\Rightarrow e^{2x} + 2\sqrt{3}e^x - 1 = 0 \Rightarrow (e^x + \sqrt{3})^2 - 3 - 1 = 0 \Rightarrow (e^x + \sqrt{3})^2 = 4$$

$$\Rightarrow e^x = -\sqrt{3} \pm 2 \Rightarrow x = \ln(-\sqrt{3}+2) \text{ only as } -\sqrt{3} - 2 < 0$$

$\therefore x = 0$ or $x = \ln(\sqrt{3}+2)$ or $x = \ln(-\sqrt{3}+2)$.

57 b)

$\cosh 3x = 5\cosh x \Rightarrow 4\cosh^3 x - 3\cosh x = 5\cosh x$

$$\Rightarrow 4\cosh^3 x - 8\cosh x = 0 \Rightarrow \cosh^3 x - 2\cosh x = 0 \Rightarrow \cosh x(\cosh^2 x - 2) = 0$$

$$\Rightarrow \cosh x = 0 \text{ or } \cosh x = \pm\sqrt{2}$$

$\cosh x = 0$ has no real solutions as $\cosh x \geq 1$. If $\cosh x = \sqrt{2}$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = \sqrt{2} \Rightarrow e^x + e^{-x} = 2\sqrt{2} \Rightarrow e^{2x} + 1 = 2\sqrt{2}e^x$$

$$\Rightarrow e^{2x} - 2\sqrt{2}e^x + 1 = 0 \Rightarrow (e^x - \sqrt{2})^2 - 2 + 1 = 0 \Rightarrow (e^x - \sqrt{2})^2 = 1$$

$$\Rightarrow e^x = \sqrt{2} \pm 1 \Rightarrow x = \ln(\sqrt{2}+1) \text{ or } \ln(\sqrt{2}-1)$$

If $\cosh x = -\sqrt{2}$ has no real solutions as $\cosh x \geq 1$. $\therefore x = \ln(\sqrt{2}+1)$ or $\ln(\sqrt{2}-1)$.

58 a)

$\sinh x = 3\ \cosh^2 x - \sinh^2 x = 1$

$$\Rightarrow \cosh^2 x - 3^2 = 1 \Rightarrow \cosh^2 x = 10 \Rightarrow \cosh x = \sqrt{10} \text{ as } \cosh x \neq -\sqrt{10}$$

58 b)

$\tanh x = \frac{\sinh x}{\cosh x}$

$$\Rightarrow \tanh x = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$$

59 a)

$\cosh x = 5$ $\cosh^2 x - \sinh^2 x = 1$

$$\Rightarrow 5^2 - \sinh^2 x = 1 \Rightarrow \sinh^2 x = 24 \Rightarrow \sinh x = \sqrt{24} = 2\sqrt{6}$$

$\sinh x \neq -\sqrt{24}$ when $x > 0$.

59 b)

$\tanh x = \frac{\sinh x}{\cosh x}$

$$\Rightarrow \tanh x = \frac{2\sqrt{6}}{5}$$

60 a)

$\tanh x = \frac{1}{3}$ $\frac{\sinh x}{\cosh x} = \frac{1}{3} \Rightarrow 3\sinh x = \cosh x \quad (1)$ $\cosh^2 x - \sinh^2 x = 1 \therefore (3\sinh x)^2 - \sinh^2 x = 1$

$$\Rightarrow 9\sinh^2 x - \sinh^2 x = 1 \Rightarrow 8\sinh^2 x = 1 \Rightarrow \sinh^2 x = \frac{1}{8}$$

$$\Rightarrow \sinh x = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} \text{ only as } x > 0$$

60 b)

Substitute into (1)

$$3 \times \frac{\sqrt{2}}{4} = \cosh x \Rightarrow \cosh x = \frac{3\sqrt{2}}{4}$$

61)

$2\cosh x + \sinh x = 2 \quad (1)$ $4\cosh^2 x - 11\sinh x = 4 \quad (2)$ (1) $2\cosh x + \sinh x = 2 \Rightarrow \sinh x = 2 - 2\cosh x$ Substitute into (2):

$$4\cosh^2 x - 11(2 - 2\cosh x) = 4 \Rightarrow 4\cosh^2 x - 22 + 22\cosh x = 4$$

$$\Rightarrow 4\cosh^2 x + 22\cosh x - 26 = 0 \Rightarrow 2\cosh^2 x + 11\cosh x - 13 = 0$$

$$\Rightarrow (2\cosh x + 13)(\cosh x - 1) = 0 \Rightarrow \cosh x = -\frac{13}{2} \text{ or } \cosh x = 1$$

$\cosh x \neq -\frac{13}{2}$ as $\cosh x \geq 1$. If $\cosh x = 1$:

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 1 \Rightarrow e^x + e^{-x} = 2 \Rightarrow e^{2x} + 1 = 2e^x$$

$$\Rightarrow e^{2x} - 2e^x + 1 = 0 \Rightarrow (e^x - 1)^2 = 0 \Rightarrow e^x = 1 \Rightarrow x = 0$$

Check in (1): $2\cosh(0) + \sinh(0) = 2 \times 1 + 0 = 2$ as required. $\therefore x = 0$.

62)

$2\cosh^2 x - \sinh x = 8 \quad (1)\ 2\cosh^2 x - 3\sinh x = 4 \quad (2)$ Rearrange (1): $\cosh^2 x = \frac{8+\sinh x}{2}$. Substitute into (2):

$$2(\frac{8+\sinh x}{2}) - 3\sinh x = 4 \Rightarrow 8 + \sinh x - 3\sinh x = 4 \Rightarrow -2\sinh x = -4 \Rightarrow \sinh x = 2$$

$$\Rightarrow \frac{e^x - e^{-x}}{2} = 2 \Rightarrow e^x - e^{-x} = 4 \Rightarrow e^{2x} - 1 = 4e^x$$

$$\Rightarrow e^{2x} - 4e^x - 1 = 0 \Rightarrow (e^x - 2)^2 - 4 - 1 = 0 \Rightarrow (e^x - 2)^2 = 5$$

$$\Rightarrow e^x = 2 \pm \sqrt{5}$$

$e^x \neq 2 - \sqrt{5}$ as $2 - \sqrt{5} < 0$ and $e^x > 0$. $\therefore x = \ln(2 + \sqrt{5})$. Check in (1):

$$2\cosh^2(\ln(2+\sqrt{5})) - \sinh(\ln(2+\sqrt{5}))$$

$$= 2(\frac{e^{\ln(2+\sqrt{5})} + e^{-\ln(2+\sqrt{5})}}{2})^2 - \frac{e^{\ln(2+\sqrt{5})} - e^{-\ln(2+\sqrt{5})}}{2}$$

$$= 2(\frac{e^{\ln(2+\sqrt{5})} + e^{\ln(2+\sqrt{5})^{-1}}}{2})^2 - \frac{e^{\ln(2+\sqrt{5})} - e^{\ln(2+\sqrt{5})^{-1}}}{2}$$

$$= 2(\frac{(2+\sqrt{5}) + (-2+\sqrt{5})}{2})^2 - \frac{(2+\sqrt{5}) - (-2+\sqrt{5})}{2}$$

$$= 2(\sqrt{5})^2 - \frac{4}{2} = 10 - 2 = 8 \text{ as required.}$$

Check in (2):

$$2\cosh^2(\ln(2+\sqrt{5})) - 3\sinh(\ln(2+\sqrt{5}))$$

$$= 2(\frac{e^{\ln(2+\sqrt{5})} + e^{-\ln(2+\sqrt{5})}}{2})^2 - 3 \times \frac{e^{\ln(2+\sqrt{5})} - e^{-\ln(2+\sqrt{5})}}{2}$$

$$= 2(\frac{e^{\ln(2+\sqrt{5})} + e^{\ln(2+\sqrt{5})^{-1}}}{2})^2 - 3 \times \frac{e^{\ln(2+\sqrt{5})} - e^{\ln(2+\sqrt{5})^{-1}}}{2}$$

$$= 2(\frac{(2+\sqrt{5}) + (-2+\sqrt{5})}{2})^2 - 3 \times \frac{(2+\sqrt{5}) - (-2+\sqrt{5})}{2}$$

$$= 2(\sqrt{5})^2 - 3 \times \frac{4}{2} = 10 - 6 = 4 \text{ as required.}$$

$\therefore x = \ln(2 + \sqrt{5})$.

63 a)

LHS: $(\sinh x + \cosh x)^2 \equiv (\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2})^2$

$$\equiv (\frac{e^x - e^{-x} + e^x + e^{-x}}{2})^2 \equiv (\frac{2e^x}{2})^2 \equiv (e^x)^2 \equiv e^{2x}$$

$$\equiv \frac{2e^{2x}}{2} \equiv \frac{e^{2x} - e^{-2x} + e^{2x} + e^{-2x}}{2}$$

$$\equiv \frac{e^{2x} - e^{-2x}}{2} + \frac{e^{2x} + e^{-2x}}{2} \equiv \sinh 2x + \cosh 2x : \text{ RHS}$$

63 b)

LHS: $(\sinh 2x - \cosh 2x)^2 \equiv (\frac{e^{2x}-e^{-2x}}{2} - \frac{e^{2x}+e^{-2x}}{2})^2$

$$\equiv (\frac{e^{2x} - e^{-2x} - e^{2x} - e^{-2x}}{2})^2 \equiv (\frac{-2e^{-2x}}{2})^2 \equiv (-e^{-2x})^2 \equiv e^{-4x} \equiv \frac{2e^{-4x}}{2}$$

$$\equiv \frac{e^{4x} + e^{-4x} - e^{4x} + e^{-4x}}{2} \equiv \frac{e^{4x} + e^{-4x}}{2} - \frac{e^{4x} - e^{-4x}}{2} \equiv \cosh 4x - \sinh 4x : \text{ RHS}$$

63 c)

LHS: $1 - \tanh^2 x \equiv 1 - \frac{\sinh^2 x}{\cosh^2 x}$

$$\equiv \frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} \equiv \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} \equiv \frac{1}{\cosh^2 x} : \text{ RHS}$$

63 d)

LHS: $\frac{1}{\tanh^2 x} - 1 \equiv \frac{1}{\frac{\sinh^2 x}{\cosh^2 x}} - 1 \equiv \frac{\cosh^2 x}{\sinh^2 x} - 1$

$$\equiv \frac{\cosh^2 x}{\sinh^2 x} - \frac{\sinh^2 x}{\sinh^2 x} \equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh^2 x} \equiv \frac{1}{\sinh^2 x} : \text{ RHS}$$

64)

LHS: $\sinh^3 x \equiv (\frac{e^x - e^{-x}}{2})^3$

$$\equiv \frac{(e^x - e^{-x})^3}{8} \equiv \frac{e^{3x} - 3e^x + 3e^{-x} - e^{-3x}}{8}$$

$$\equiv \frac{e^{3x} - e^{-3x}}{8} + \frac{-3e^x + 3e^{-x}}{8}$$

$$\equiv \frac{1}{4} \times \frac{e^{3x} - e^{-3x}}{2} - \frac{3}{4} \times \frac{e^x - e^{-x}}{2} \equiv \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x : \text{ RHS}$$

65)

LHS: $\cosh 3x - 3\cosh x \equiv \frac{e^{3x}+e^{-3x}}{2} - 3 \times \frac{e^x + e^{-x}}{2}$

$$\equiv \frac{(e^x + e^{-x})(e^{2x} + e^{-2x} - 1)}{2} - 3 \times \frac{e^x + e^{-x}}{2}$$

$$\equiv \frac{e^x + e^{-x}}{2} \times (e^{2x} + e^{-2x} - 1) - 3 \times \frac{e^x + e^{-x}}{2}$$

$$\equiv \frac{e^x + e^{-x}}{2}(e^{2x} + e^{-2x} - 1 - 3) \equiv \frac{e^x + e^{-x}}{2}(e^{2x} + e^{-2x} - 4)$$

$$\equiv \frac{e^x + e^{-x}}{2}(2(\frac{e^{2x} + e^{-2x}}{2} - 2))$$

$$\equiv 2 \times \frac{e^x + e^{-x}}{2}(\frac{e^{2x} + e^{-2x}}{2} - 2) \equiv 2\cosh x(\cosh 2x - 2) : \text{ RHS}$$

66)

LHS: $\cosh 2x - 2\sinh^2 x$

$$\equiv \frac{e^{2x}+e^{-2x}}{2} - 2(\frac{e^{x}-e^{-x}}{2})^2$$

$$\equiv \frac{e^{2x}+e^{-2x}}{2} - 2 \times \frac{e^{2x}-2+e^{-2x}}{4}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{2} - \frac{e^{2x}+e^{-2x}-2}{2}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{2} - \frac{e^{2x}+e^{-2x}}{2} + 1 \equiv 1 : \text{ RHS}$$

67)

RHS: $\sinh(\frac{x}{2})(6\cosh(\frac{x}{2}) - 7)$

$$\equiv \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2} \times (6 \times \frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2} - 7)$$

$$\equiv \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2} \times 6 \times \frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2} - 7 \times \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$$

$$\equiv \frac{3}{2}(e^{\frac{x}{2}}-e^{-\frac{x}{2}})(e^{\frac{x}{2}}+e^{-\frac{x}{2}}) - 7 \times \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$$

$$\equiv \frac{3}{2}(e^{x}-e^{-x}) - 7 \times \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2} \equiv 3 \times \frac{e^{x}-e^{-x}}{2} - 7 \times \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$$

$$\equiv 3\sinh x - 7\sinh(\frac{x}{2}) : \text{ LHS}$$

68)

RHS: $4\sinh(\frac{x}{4})\cosh(\frac{x}{4})\cosh(\frac{x}{2})$

$$\equiv 4(\frac{e^{\frac{x}{4}}-e^{-\frac{x}{4}}}{2})(\frac{e^{\frac{x}{4}}+e^{-\frac{x}{4}}}{2})(\frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2})$$

$$\equiv \frac{1}{2}(e^{\frac{x}{4}}-e^{-\frac{x}{4}})(e^{\frac{x}{4}}+e^{-\frac{x}{4}})(e^{\frac{x}{2}}+e^{-\frac{x}{2}})$$

$$\equiv \frac{1}{2}(e^{\frac{x}{2}}-e^{-\frac{x}{2}})(e^{\frac{x}{2}}+e^{-\frac{x}{2}}) \equiv \frac{1}{2}(e^{x}-e^{-x})$$

$$\equiv \frac{e^{x}-e^{-x}}{2} \equiv \sinh x : \text{ LHS}$$

69)

LHS: $\cosh^2(\frac{x}{3}) - 4\sinh^2(\frac{x}{6})\cosh^2(\frac{x}{6})$

$$\equiv (\frac{e^{\frac{x}{3}}+e^{-\frac{x}{3}}}{2})^2 - 4(\frac{e^{\frac{x}{6}}-e^{-\frac{x}{6}}}{2})^2(\frac{e^{\frac{x}{6}}+e^{-\frac{x}{6}}}{2})^2$$

$$\equiv \frac{1}{4}(e^{\frac{x}{3}}+e^{-\frac{x}{3}})^2 - \frac{1}{4}(e^{\frac{x}{6}}-e^{-\frac{x}{6}})^2(e^{\frac{x}{6}}+e^{-\frac{x}{6}})^2$$

$$\equiv \frac{1}{4}(e^{\frac{x}{3}}+e^{-\frac{x}{3}})^2 - \frac{1}{4}((e^{\frac{x}{6}}-e^{-\frac{x}{6}})(e^{\frac{x}{6}}+e^{-\frac{x}{6}}))^2$$

$$\equiv \frac{1}{4}(e^{\frac{x}{3}}+e^{-\frac{x}{3}})^2-\frac{1}{4}(e^{\frac{x}{3}}-e^{-\frac{x}{3}})^2$$

$$\equiv \frac{1}{4}((e^{\frac{x}{3}}+e^{-\frac{x}{3}})^2-(e^{\frac{x}{3}}-e^{-\frac{x}{3}})^2)$$

$$\equiv \frac{1}{4}(e^{\frac{2x}{3}}+2+e^{-\frac{2x}{3}}-e^{\frac{2x}{3}}+2-e^{-\frac{2x}{3}}) \equiv \frac{1}{4}(4) \equiv 1: \text{ RHS}$$

70)

RHS: $4\cosh^4(\frac{x}{4})-\sinh^2(\frac{x}{2})\tanh^2(\frac{x}{4})$

$$\equiv 4(\frac{e^{\frac{x}{4}}+e^{-\frac{x}{4}}}{2})^4-(\frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2})^2(\frac{e^{\frac{x}{4}}-e^{-\frac{x}{4}}}{e^{\frac{x}{4}}+e^{-\frac{x}{4}}})^2$$

$$\equiv \frac{1}{4}(e^{\frac{x}{4}}+e^{-\frac{x}{4}})^4-\frac{1}{4}(e^{\frac{x}{2}}-e^{-\frac{x}{2}})^2\frac{(e^{\frac{x}{4}}-e^{-\frac{x}{4}})^2}{(e^{\frac{x}{4}}+e^{-\frac{x}{4}})^2}$$

$$\equiv \frac{1}{4}(e^{\frac{x}{2}}+2+e^{-\frac{x}{2}})^2-\frac{1}{4}\frac{(e^{\frac{x}{2}}-e^{-\frac{x}{2}})^2(e^{\frac{x}{2}}-2+e^{-\frac{x}{2}})}{e^{\frac{x}{2}}+2+e^{-\frac{x}{2}}}$$

$$\equiv \frac{1}{4}e^x+e^{\frac{x}{2}}+\frac{3}{2}+e^{-\frac{x}{2}}+\frac{1}{4}e^{-x}-\frac{1}{4}\frac{(e^{\frac{x}{2}}-e^{-\frac{x}{2}})^2(e^{\frac{x}{2}}-2+e^{-\frac{x}{2}})}{e^{\frac{x}{2}}+2+e^{-\frac{x}{2}}}$$

$$\equiv \frac{1}{4}e^x+e^{\frac{x}{2}}+\frac{3}{2}+e^{-\frac{x}{2}}+\frac{1}{4}e^{-x}-\frac{1}{4}\frac{(e^{\frac{x}{2}}+e^{-\frac{x}{2}})^2(e^{\frac{x}{2}}-e^{-\frac{x}{2}})^2(e^{\frac{x}{2}}-e^{-\frac{x}{2}})^2}{(e^{\frac{x}{2}}+e^{-\frac{x}{2}})^2}$$

$$\equiv \frac{1}{4}e^x+e^{\frac{x}{2}}+\frac{3}{2}+e^{-\frac{x}{2}}+\frac{1}{4}e^{-x}-\frac{1}{4}(e^{\frac{x}{2}}-e^{-\frac{x}{2}})^4$$

$$\equiv \frac{1}{4}e^x+e^{\frac{x}{2}}+\frac{3}{2}+e^{-\frac{x}{2}}+\frac{1}{4}e^{-x}-\frac{1}{4}(e^x-4e^{\frac{x}{2}}+6-4e^{-\frac{x}{2}}+e^{-x})$$

$$\equiv \frac{1}{4}e^x+e^{\frac{x}{2}}+\frac{3}{2}+e^{-\frac{x}{2}}+\frac{1}{4}e^{-x}-\frac{1}{4}e^x+e^{\frac{x}{2}}-\frac{3}{2}+e^{-\frac{x}{2}}-\frac{1}{4}e^{-x}$$

$$\equiv 2e^{\frac{x}{2}}+2e^{-\frac{x}{2}}=4(\frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2}) \equiv 4\cosh(\frac{x}{2}): \text{ LHS}$$